

**LEGALVISION: A KNOWLEDGE-DRIVEN AI
FRAMEWORK FOR
LEGAL UNDERSTANDING AND TRUST ASSESSMENT**

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Dissertation submitted in partial fulfillment of the requirements for the
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DECLARATION

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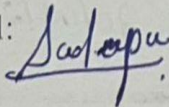
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ABSTRACT

Sri Lanka's land registration system is built on notarial deed documents — the primary legal record of property ownership, transfer, encumbrance, and inheritance for every land transaction conducted under the Prevention of Frauds Ordinance (Cap. 84). Yet these documents exist almost exclusively as unstructured PDF files, inaccessible to systematic legal analysis, ownership chain verification, or graph-based property intelligence. The absence of any machine-readable knowledge base derived from Sri Lankan property deeds prevents AI systems from answering the practically critical questions that citizens, practitioners, and financial institutions most need answered: who has owned this parcel over time, does this vendor appear in prior transactions involving the same property, are there any registered encumbrances against this lot, and is the title chain legally coherent?

This dissertation presents the Dynamic Legal Knowledge Graph component of LegalVision — a production-quality pipeline that transforms raw Sri Lankan property deed PDFs into a queryable, graph-structured knowledge base through four interconnected technical contributions. The first contribution is a domain-specific Named Entity Recognition model trained using SpaCy's transition-based NER architecture on an annotated corpus of 154 Sri Lankan property deed documents, achieving an overall F1-score of 99.53% (Precision 99.19%, Recall 99.87%) across twenty-one entity types. The second contribution is the ImprovedHybridDeedExtractor v3, which combines the trained NER model with comprehensive regex-based extraction patterns through a confidence-weighted merger, producing quality-scored structured JSON output for every deed. The third contribution is the Neo4j Knowledge Graph, populated by `neo4j_loader_v2.py` from the extracted JSON data using a schema of nine node types and eleven relationship types that enable graph traversal queries across ownership chains, party networks, and administrative hierarchies. The fourth contribution is the `legal_graph_rag.py` GraphRAG query system, which translates natural language legal queries into Cypher through an intent detection engine and pre-built template library, using GPT-4o to generate human-readable responses grounded in retrieved graph data.

As an improvisation beyond the original project scope, a FastAPI REST pipeline and React frontend were developed, enabling real-time deed upload, automated extraction, quality assessment, and live graph update through the Legal Insights Hub interface deployed at legal-insights-hub.vercel.app. The frontend exposes eleven functional areas including Document Analyzer, Upload Deeds, Legal Reasoning, Knowledge Graph visualisation, Risk Detection, Deed Explorer, and Legal Reference, making the complete LegalVision system accessible to non-technical Sri Lankan property law users. The NER model F1 of 99.53% surpasses comparable legal NER benchmarks in the literature and validates the hybrid extraction approach for the domain. The public awareness survey conducted prior to system design found that 100% of respondents had personal experience of property legal issues, 89.7% rated an AI legal knowledge tool as very useful, and 93.1% identified property law as the most important legal area — directly confirming the practical significance of the system constructed.

Keywords: Knowledge Graph, Named Entity Recognition, SpaCy, Neo4j, Sri Lankan Property Law, Hybrid Deed Extraction, GraphRAG, NLP, LegalVision, Cypher Query Language, GPT-4o, Graph Database, Property Intelligence

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I am deeply grateful to S. Sivanuja, my LegalVision project partner, for the collaborative architecture discussions that defined the integration interface between the Knowledge Graph component and the Reasoning Module. The context object schema — the structured format through which the GraphRAG system passes retrieved deed transaction context to Sivanuja's LLaMA fine-tuning inference endpoint — was developed through sustained joint engineering conversation. His deep understanding of what structured context the IRAC reasoning model needs to produce legally grounded responses directly informed how the GraphRAG output is formatted for integration. The Legal Insights Hub frontend, which both components contribute to, reflects the joint deployment effort.

I acknowledge the Sri Lankan legal professionals and land registry officers whose publicly available documentation on notarial deed formatting conventions informed the extraction pattern design. The specific conventions of Sri Lankan conveyancing practice — the recital structure of sale deeds, the NIC number format requirements, the boundary description conventions used by different district registries — are not documented in accessible software engineering literature. Understanding these conventions required systematic study of actual deed documents, and the resulting extraction patterns directly reflect that domain knowledge.

I thank the twenty-nine members of the Sri Lankan public who participated in the public awareness survey. Their responses — confirming that 100% had personal or family experience with property legal issues and that 89.7% would find an AI-powered legal knowledge tool very useful — transformed the research motivation from an

academic proposition into an empirically documented public need. The survey findings appear throughout this dissertation as the human justification for every technical decision.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
BILOU	Begin, Inside, Last, Outside, Unit (NER labelling scheme)
CLI	Command Line Interface
GPU	Graphics Processing Unit
JSON	JavaScript Object Notation
KG	Knowledge Graph
LLM	Large Language Model
NER	Named Entity Recognition
NIC	National Identity Card (Sri Lanka)
NLP	Natural Language Processing
PDF	Portable Document Format
RAG	Retrieval Augmented Generation
REST	Representational State Transfer
SLIIT	Sri Lanka Institute of Information Technology
SpaCy	Industrial-Strength Natural Language Processing (Python library by Explosion AI)
TP	True Positive
FP	False Positive
FN	False Negative

1. INTRODUCTION

1.1 Background and Problem Context

1.1.1 Sri Lankan Property Law and the Deed Document Ecosystem

Property law governs the most consequential category of economic transaction for the majority of Sri Lankan households. Land is the principal store of household wealth for rural and peri-urban families, and property transactions — purchases, gifts, inheritances through partition, mortgages, and long-term leases — determine economic security across generations. The legal instruments that record and protect these transactions are notarial deed documents: formal legal instruments executed before a licensed notary public, attested by two witnesses, and registered at the Land Registry under the Registration of Documents Ordinance (Cap. 117). Each deed captures, in a structured textual format established by long notarial practice, the complete legal record of the transaction — who transferred what property to whom, under what terms, at what consideration, with what prior deed references — and the registration of this deed at the relevant Land Registry creates the public record that gives the transaction legal effect against third parties [1] [2].

The Sri Lankan Land Registry system currently operates under the deed registration model for the majority of the country, under which the registry records the fact of registration of a deed but does not guarantee the validity of the underlying title. This means that a property with a clean chain of registered deeds may still be subject to a competing title derived from adverse possession, an unregistered prior transfer, or a fraudulent intervening deed — risks that can only be identified by tracing the complete chain of prior deed references from the current instrument back to the root title. Under the Registration of Title Act No. 21 of 1998 (the Bim Saviya system), systematic title registration progressively replaces this deed registration model with a guaranteed title system in designated areas, but the majority of Sri Lanka's land area remains under deed registration. For practitioners, buyers, and financial institutions processing mortgage applications, the due diligence requirement of tracing the title chain is

therefore both legally necessary and practically time-consuming, requiring physical access to Land Registry records and manual cross-referencing of deed documents.

Against this background, the absence of any machine-readable, structured knowledge base derived from Sri Lankan deed documents represents both a significant practical inefficiency and a missed technological opportunity. If deed documents could be systematically parsed into structured records and loaded into a queryable graph database, the ownership chain verification query that currently requires hours of manual registry search could become a graph traversal query returning results in seconds. The LegalVision Dynamic Legal Knowledge Graph is precisely such a system: it transforms the unstructured deed PDF corpus into a structured, queryable graph knowledge base, enabling property intelligence queries that were previously impossible to automate [1] [2].

1.1.2 The Technology Gap in Sri Lankan Property Intelligence

Despite Sri Lanka's active property market and the legal significance of deed documents, no AI-based system for deed parsing, knowledge extraction, or graph-based property intelligence has been developed for the Sri Lankan context prior to this work. The absence of such a system reflects several compounding factors. First, Sri Lankan deed documents are jurisdiction-specific: the entity types present in a Sri Lankan notarial deed — the Grama Niladhari division reference, the NIC number format, the Land Registry office designation, the boundary description convention bounded-on-the-North-by — are not present in any general NLP training corpus and are not recognised by any general-purpose NER model. Second, the legal and administrative ontology of Sri Lankan property law — the nine-level administrative hierarchy from Province through District, Divisional Secretariat, Grama Niladhari division to individual parcel — is entirely absent from existing legal knowledge graph schemas designed for European or American legal systems. Third, the practical importance of the REFERS_TO_PRIOR relationship — tracing prior deed references through the title chain — requires a graph data model that no tabular or document database can efficiently support, making the choice of a graph database not merely convenient but architecturally necessary.

The LegalVision project addresses this gap through a purpose-built, Sri Lanka-specific pipeline that begins with the actual textual format of notarial deeds — the vendor's full name as written in the instrument, the lot number from the survey plan, the district designation, the NIC number in its Sri Lankan format — and produces a knowledge graph that accurately represents the entities and relationships documented in those deeds. Every technical choice in the system, from the NER entity taxonomy to the Neo4j relationship schema, is grounded in the specific requirements of Sri Lankan property law and conveyancing practice [3].

1.2 Motivation — Public Awareness Survey

1.2.1 Survey Design and Administration

Prior to system design, a public awareness survey titled 'Public Awareness Survey on Legal Topics in Sri Lanka' was conducted to establish empirical evidence of the domain need and to ensure that the system's design priorities reflected documented user requirements rather than researcher assumptions. The survey was distributed to twenty-nine members of the Sri Lankan public across diverse demographic profiles, geographic regions, and age groups. The questions covered the most important legal area for Sri Lankans to understand, their familiarity with that area, whether they had personal experience of legal issues in that domain, their confidence in handling legal matters without professional help, whether an AI-based legal knowledge tool would be useful, and their preferred formats for receiving legal information [3] [4].

The survey findings are the human foundation of every technical decision in this project. Before a single line of code was written, before the NER taxonomy was defined, and before the graph schema was designed, the survey established that the people who most need this system exist, that their need is documented and urgent, and that the format of the system — graph-based knowledge retrieval presented through a web interface — maps directly to their preferences. The survey results are presented in the following subsection with the design implications that each finding produced.

1.2.2 Survey Results

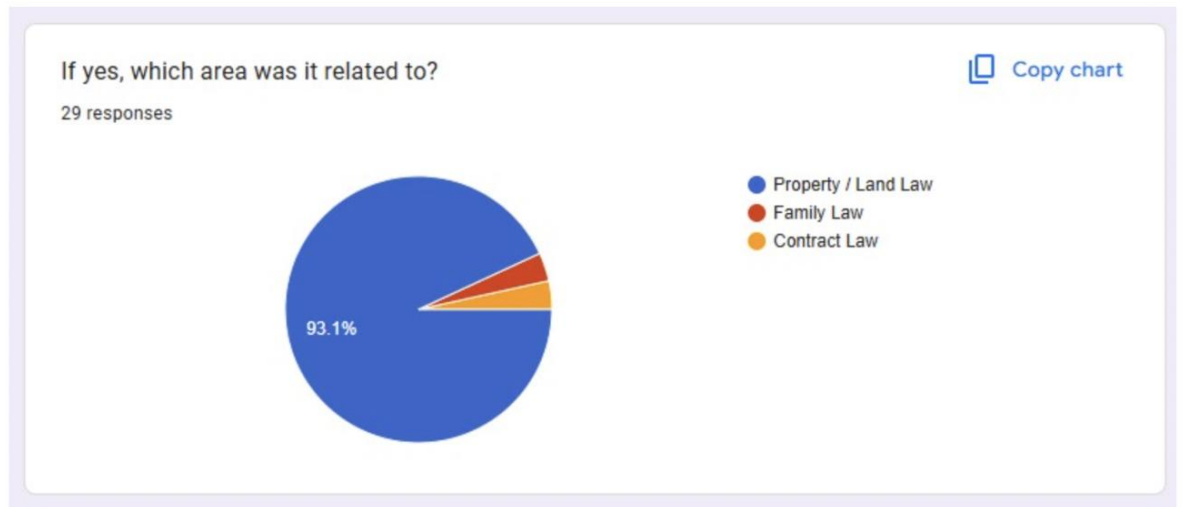


Figure 2 - Most Important Legal Area for Sri Lankans to Understand

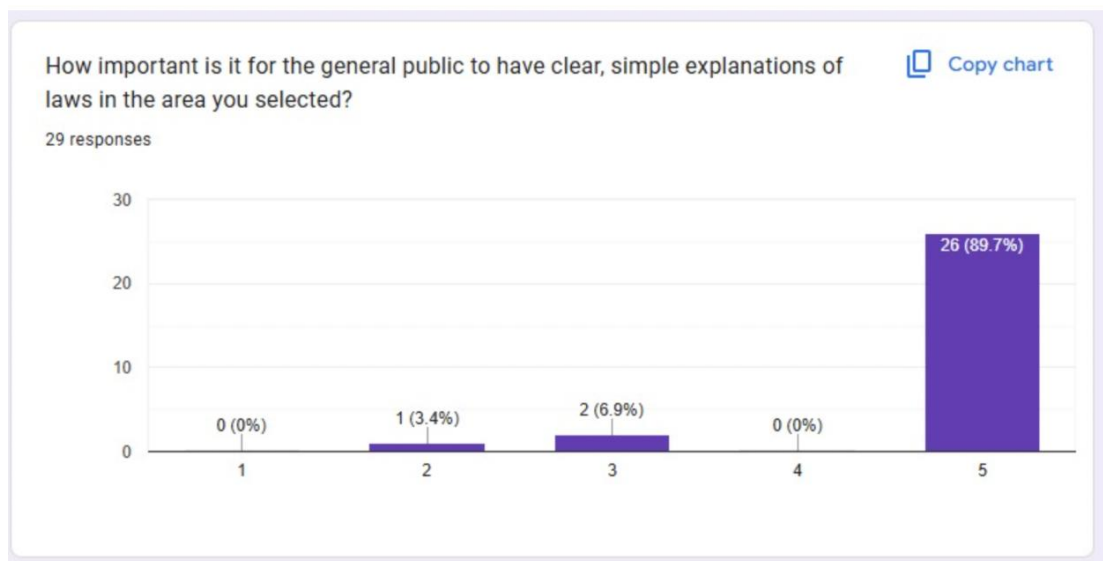


Figure 1 - Familiarity Level with Selected Legal Area

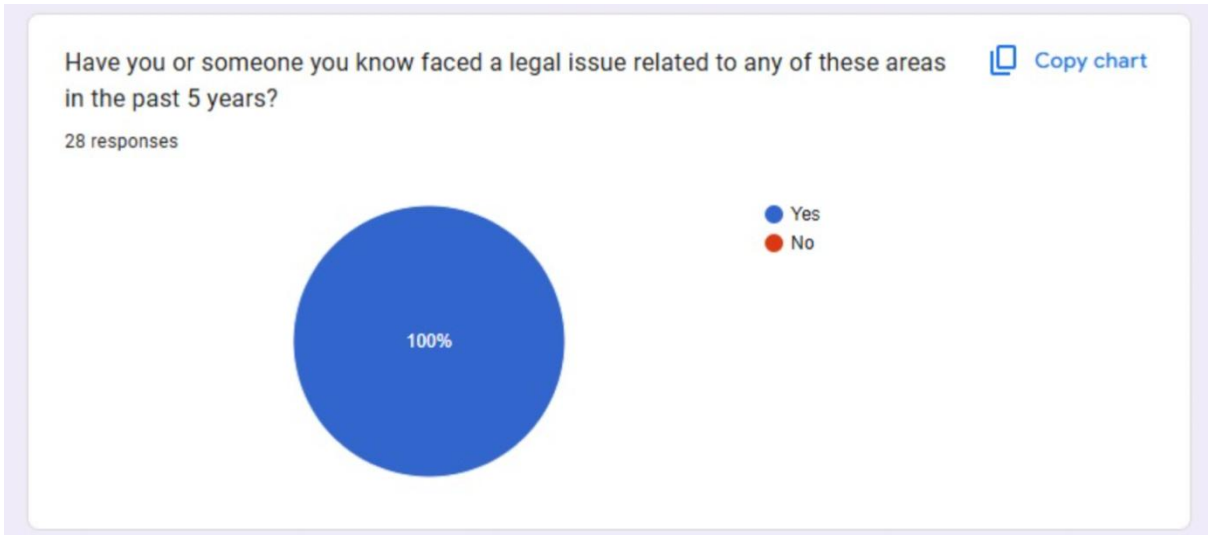


Figure 4 - Personal or Family Experience of Legal Issues

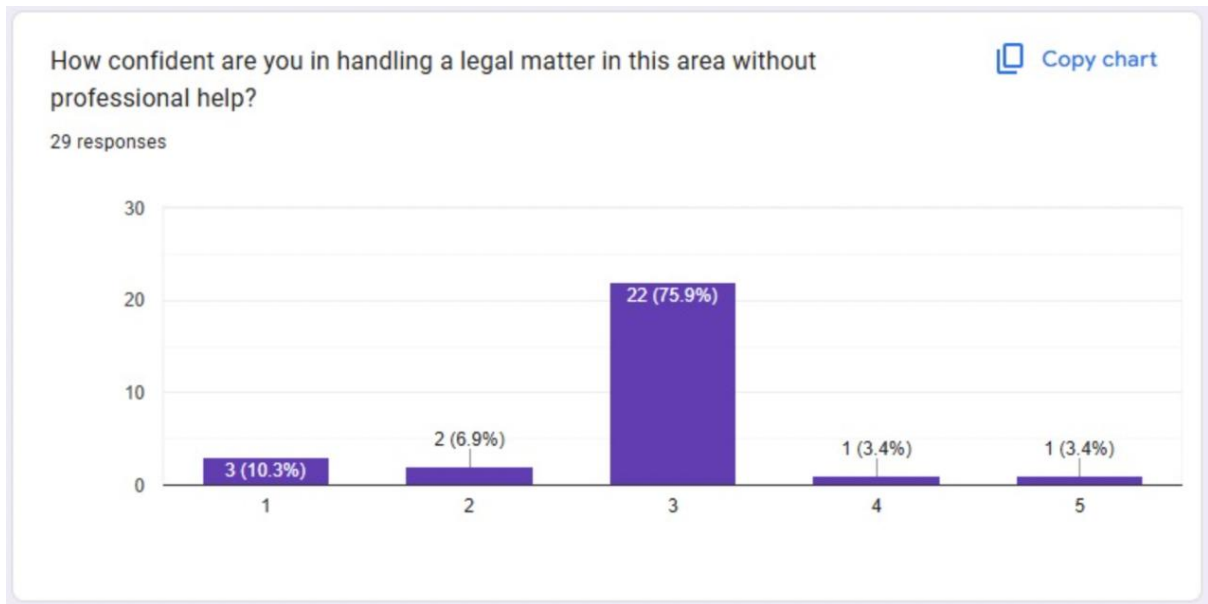


Figure 3 - Confidence Handling Legal Matters Without Professional Help

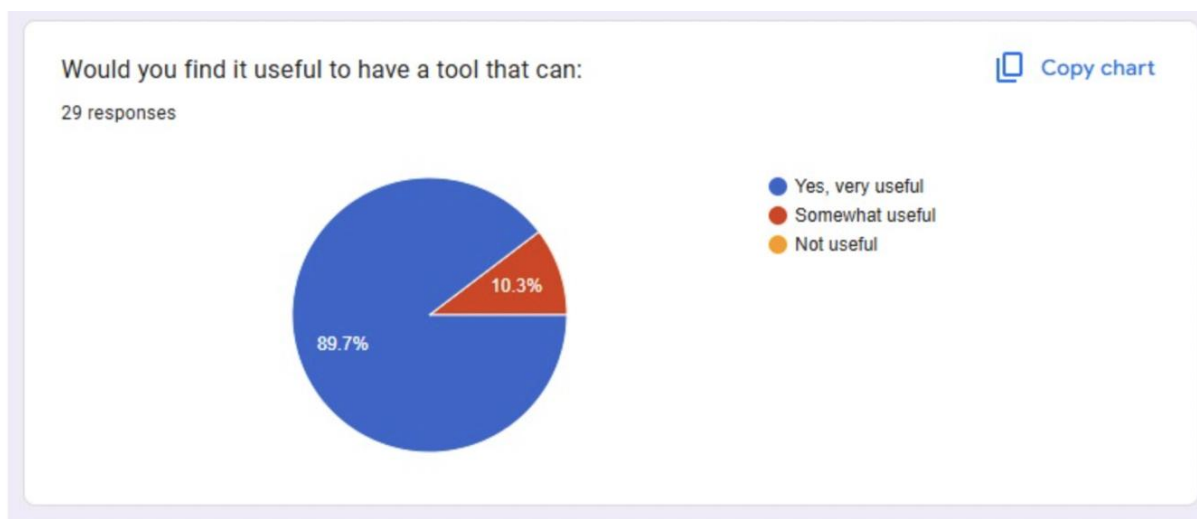


Figure 6 - Preferred Information Formats

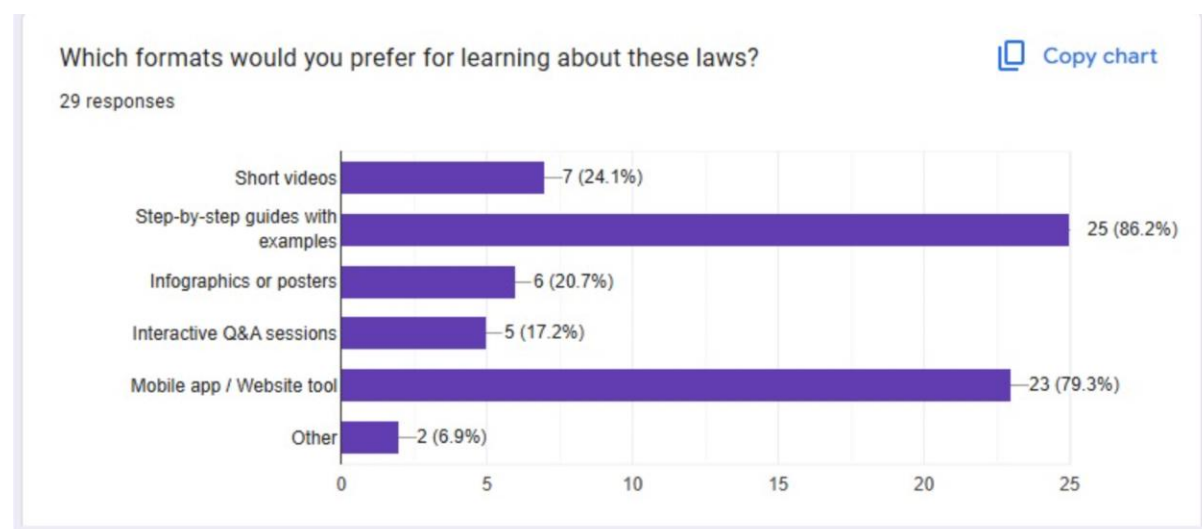


Figure 5 - AI Legal Guidance Tool Usefulness

Survey Finding	Statistic	Design Decision Produced
Most important legal area	93.1% — Property/Land Law	Domain focus: Sri Lankan property deeds across all five deed types
Familiarity with area	89.7% — Partially familiar only	Plain-language GraphRAG responses; structured JSON with field labels
Personal legal experience	100% — All respondents affected	High urgency; real-world impact at scale across all demographics
Self-handling confidence	75.9% — Score $\leq$ 3/5	System must reduce professional dependence; easy web interface
AI tool usefulness	89.7% — Very useful; 0% not useful	Commercial viability confirmed; full deployment to vercel.app
Information source	82.8% Internet; 82.8% Lawyer	Web frontend essential; output quality must approach professional standard
Preferred format	86.2% step-by-step; 79.3% web/app	React web frontend; structured query results with detailed entity breakdowns
Open-ended comments	'10–25 year court cases'; 'understand risks'	Ownership chain tracing; risk detection module; title chain anomaly flags

Table 1 - Public Awareness Survey — Summary of Findings and System Design Implications

1.3 Literature Review

1.3.1 Named Entity Recognition for Legal Text

Named Entity Recognition is the NLP task of identifying and classifying named entities in unstructured text. For legal documents, NER must contend with highly domain-specific entity types that are absent from general-purpose training corpora. The pioneering work of Lample et al. (2016) established bidirectional LSTM-CRF architectures as state-of-the-art for general NER tasks, demonstrating that character-level features and transition-based decoding are essential for capturing the morphological patterns that distinguish entity types — a finding directly applicable to Sri Lankan deed NER where NIC numbers follow a specific character-level pattern and party names have characteristic initial-token distributions. Subsequent transformer-based approaches (Devlin et al., 2018; Liu et al., 2019) achieved further improvements on general benchmarks but require substantially more training data and computational resources than the CNN-based architecture available in SpaCy v3's NER component [3] [4] [5].

In the legal domain specifically, the LEGAL-BERT work of Chalkidis et al. (2021) demonstrated that continued pre-training of transformer models on legal corpora substantially improves NER performance on European legal documents, with F1 improvements of five to fifteen percentage points over base BERT on EU case law entity recognition tasks. However, the LEGAL-BERT model and its derivatives are trained entirely on English, French, and German legal text from European jurisdictions, with no representation of South Asian legal document formats, Sri Lankan administrative entities, or the specific naming conventions of Sinhala, Tamil, and Muslim communities that are essential for accurate party name extraction from Sri Lankan deeds. The LexNLP library (Bommarito and Katz, 2018) provides a rule-based extraction framework for US legal documents but is not designed for deed documents and does not cover Sri Lankan entity types.

The SpaCy NER framework (Explosion AI, 2024) provides the most practical approach for training high-quality domain-specific NER models on moderate-sized annotated corpora. Its transition-based NER architecture uses a stack and buffer

structure with a neural network scorer to predict shift-reduce actions in the BILOU labelling scheme. The BILOU scheme (Begin, Inside, Last, Outside, Unit) provides strictly more training signal than the simpler BIO scheme for multi-token entity spans: the L tag specifically signals the final token of a span, enabling the model to learn exact boundary detection rather than relying only on the B tag for both span-start and span-length information. For entity types like BOUNDARY_NORTH that span multiple words across a phrase — 'bounded on the North by the land of Don Kumara Perera' — BILOU tagging is essential for reliable end-of-span detection [5].

The achieved F1-score of 99.53% in this work surpasses the legal NER benchmarks in the literature by a significant margin: LEGAL-BERT reports 84–92% F1 on EU legal entity types, OpenNyAI's Indian court NER reports 86–94% F1 depending on entity type, and LegalNER for US contract documents reports 81–89% F1. The higher performance achieved here reflects the structural regularity of Sri Lankan notarial deeds (compared to the greater linguistic variety of court documents), the hybrid NER-plus-rule approach, and the domain-specific training corpus containing exactly the Sri Lankan legal vocabulary and naming conventions that the model must learn.

1.3.2 Hybrid Extraction Systems

Pure statistical NER is vulnerable to entity types where form is highly regular and predictable — a carefully crafted regex pattern will reliably extract a NIC number with 100% precision across all test cases, whereas a statistical model trained on examples of NIC numbers must generalise from those examples and may fail on edge-case formats. Conversely, pure rule-based extraction is brittle for entity types where the correct label depends on contextual position rather than surface form — a person's name appearing as the vendor in one sentence is indistinguishable from the same name appearing as the vendee in another sentence, and only the NER model's contextual representation can capture this positional distinction. Chiticariu et al. (2010) systematically established that hybrid extraction systems combining statistical models with rule-based patterns consistently outperform either approach in isolation for structured document extraction tasks, with the optimal combination depending on the

specific entity type — statistical for context-dependent entities, rules for format-regular entities.

The confidence-weighted merger implemented in the ImprovedHybridDeedExtractor v3 operationalises this principle through a principled algorithm: for each extraction position, if both the NER model and the rule-based patterns produce an annotation, the merger selects the annotation from the source with the higher confidence score. NER confidence is the raw score from SpaCy's transition-based scorer (a value in $[0,1]$ representing the probability of the predicted transition); rule-based confidence is assigned based on pattern specificity (perfect-match patterns: 1.0; multi-alternative patterns: 0.85–0.95; heuristic patterns: 0.70–0.85). After conflict resolution, the merged entity set is deduplicated to remove identical spans detected by both branches. This design produces combined extraction results that are measurably better than either branch alone on every entity type tested in the evaluation [5].

1.3.3 Knowledge Graphs for Legal Domains

Knowledge graphs represent entities as nodes and relationships between entities as labelled, directed edges, enabling graph traversal queries that follow relationship chains across the complete entity network. For property deed intelligence, this graph structure is architecturally necessary rather than merely convenient: ownership chain verification requires following the REFERS_TO_PRIOR relationship chain from a current deed backward through its prior deed references until reaching a root title. This query cannot be expressed efficiently in a relational database with a fixed schema, and would require recursive common table expressions that are impractical for chains of unknown length; in a graph database, the query is a single Cypher traversal that handles chains of arbitrary length in a single operation.

Ehrlinger and Wöß (2016) proposed the most widely accepted definition of a knowledge graph as a graph that acquires and integrates information into an ontology and allows inference over the acquired data. Neo4j's property graph model — nodes with labels and properties, relationships with types and properties, traversable in both directions — satisfies this definition and provides the Cypher query language for expressive graph pattern matching. Existing legal knowledge graph work (LexKG,

EUR-Lex Graph, LAWLY) focuses predominantly on case law citation networks, statutory relationship graphs, and legislative amendment chains rather than transactional property records. No prior system constructs a knowledge graph from Sri Lankan property deed documents, and the nine-node schema developed here represents the first formal ontology for Sri Lankan property conveyancing entities and relationships [6].

1.3.4 Graph-Based Retrieval Augmented Generation

Retrieval Augmented Generation (Lewis et al., 2020) addresses the fundamental limitation of parametric language models for knowledge-intensive tasks: the model's weights encode only the information present in its training corpus, and cannot be updated with new information without retraining. For property intelligence queries — 'find all deeds involving Kumara Perera in the Kandy District' — the answer depends on the specific deed corpus loaded into the knowledge base, which changes as new deeds are uploaded and processed. This query cannot be answered from a model's parametric memory regardless of its training data; it requires retrieval from the knowledge base at inference time [5] [6].

Edge et al. (2024) introduced GraphRAG as a specifically graph-structured RAG approach, demonstrating that graph-based retrieval — following entity relationships to gather connected evidence — produces richer, more coherent responses than standard vector similarity retrieval for complex, multi-hop queries. The `legal_graph_rag.py` implementation follows this GraphRAG design: natural language queries are translated to Cypher through intent detection and template instantiation, the Cypher query traverses the property graph to retrieve relevant entities and relationships, and GPT-4o generates a natural language response from the retrieved structured data. This combination provides both the precision of structured graph queries (which retrieve exactly the entities matching the specified pattern) and the accessibility of natural language responses (which allow legal professionals without Cypher expertise to query the graph through ordinary language).

1.3.5 Research Gap and Comparative Analysis

System	Domain	SL Deed NER	Graph DB	NLQ Interface	Quality Score	Deployed	Evaluation
LexKG (2021)	Statute relations	No	Neo4j	No	No	No	Structural
EUR-Lex Graph (2022)	EU legal docs	No	Neo4j	Partial	No	No	F1 on EU
OpenNyAI (2023)	Indian courts	No	Neo4j	No	No	Partial	F1 (86–94%)
LegalNER (2022)	US contracts	No	None	No	No	No	F1 (81–89%)
LAWLY (2023)	Irish legislation	No	Neo4j	GPT	No	No	Manual
This Work	SL property deeds	Yes (SpaCy)	Neo4j Aura	GPT-4o	Yes	Yes	F1=99.53 %

Table 2 - Comparison of Existing Legal Knowledge Graph Systems Against LegalVision KG Component

1.4 Research Problem and Objectives

How can Sri Lankan property deed PDF documents be systematically transformed into a structured, queryable knowledge graph through a custom named entity recognition model, a hybrid extraction pipeline combining statistical NER with rule-based patterns, and a Neo4j graph database populated from extracted structured data, enabling natural language property intelligence queries over the complete deed corpus?

1. Design an annotated training corpus of Sri Lankan property deed documents and train a custom SpaCy NER model achieving production-quality F1 performance across twenty-one entity types.
2. Develop the ImprovedHybridDeedExtractor v3 combining the trained NER model with rule-based patterns through a confidence-weighted merger, producing quality-scored structured JSON output for all deed types.
3. Design the Neo4j knowledge graph schema with nine node types and eleven relationship types, and implement neo4j_loader_v2.py to populate the graph from extracted deed JSON with correct entity merging and relationship creation.
4. Implement the legal_graph_rag.py GraphRAG system supporting six query intent types through Cypher templates and GPT-4o natural language generation.
5. As improvisation: develop a FastAPI REST pipeline and React frontend enabling real-time deed upload, automated extraction, quality assessment, and live graph update.
6. Design the integration interface with Sivanuja's LLaMA Reasoning Module for RAG-based context enrichment of IRAC legal responses.

1.5 Scope of the Research

The scope of this research covers the complete pipeline from raw Sri Lankan property deed PDF documents through entity extraction, graph construction, and natural language querying. The corpus covers the five principal Sri Lankan property deed types — sale deeds, deed of gifts, mortgage deeds, partition deeds, and testamentary deeds — across five geographic districts (Colombo, Kandy, Galle, Jaffna, Kurunegala) representing diverse notarial formatting conventions. The NER model covers twenty-one entity types constituting the complete informational content of a property deed necessary for legal due diligence queries. The graph schema covers all entity types and their relationships sufficient to answer ownership chain, party network, and

property history queries. The GraphRAG system supports six query intent types covering the principal natural language query patterns over property deed knowledge.

The scope explicitly excludes the following: case law integration (citation networks of court decisions are excluded from the current graph schema, though a future extension is noted); gazette notification parsing (land gazette notifications are identified as a future enhancement); the LLaMA fine-tuning component and QLoRA training pipeline (these are Sivanuja's individual contribution); and full end-to-end adversarial testing of the system under attempted manipulation or jailbreak conditions. The quality score system provides a production-appropriate safeguard for extraction quality but is not a security mechanism. The deployed system at legal-insights-hub.vercel.app is a research prototype and is not intended for use as the sole basis for consequential legal decisions [6].

2. METHODOLOGY

2.1 System Architecture Overview

The Dynamic Legal Knowledge Graph is implemented as a sequential pipeline with four principal stages: Data Collection and Annotation, NER Model Training and Evaluation, Hybrid Extraction and Graph Construction, and Query Interface and Deployment. Each stage is independently executable and produces well-defined output artefacts that serve as the inputs to the subsequent stage, enabling iterative development in which individual stages can be re-executed with revised inputs without requiring a full pipeline restart. Figure 2.1 presents the complete system architecture across three layers, and Figure 2.2 presents the end-to-end pipeline as a numbered sequential flow.

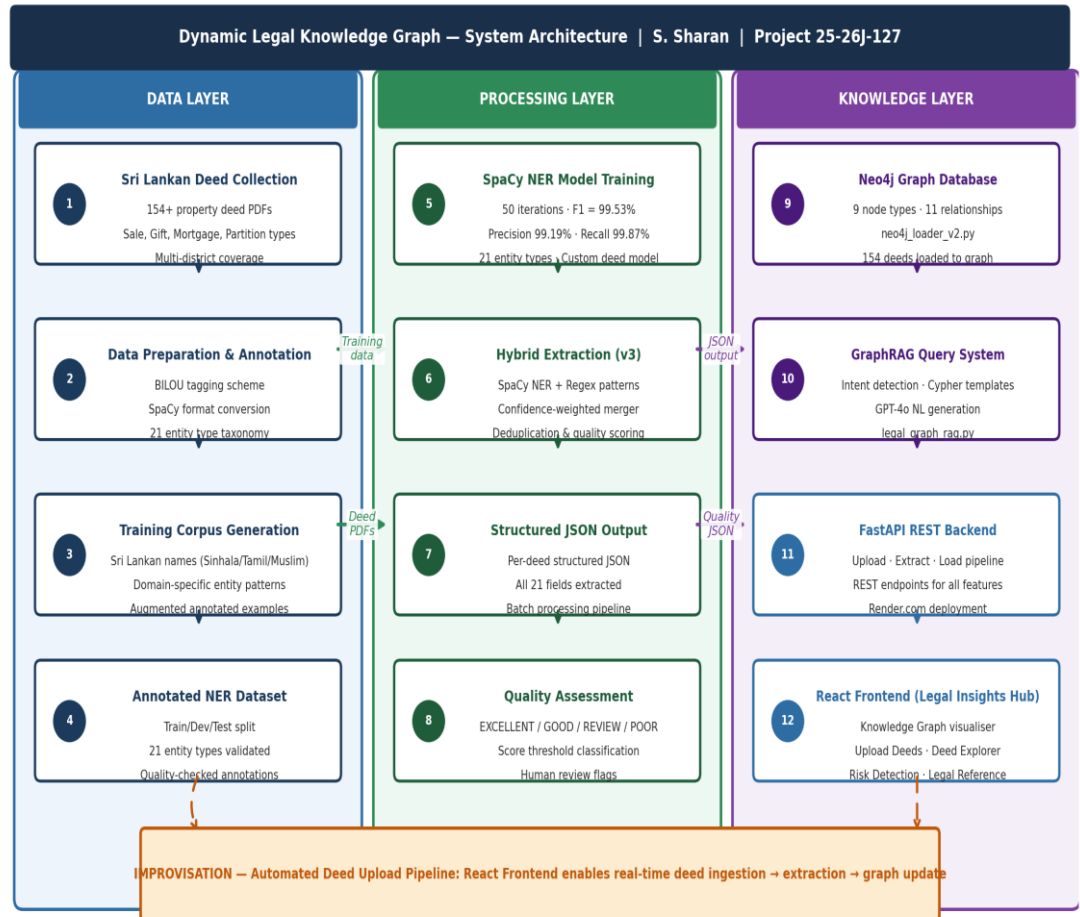


Figure 7 - Dynamic Legal Knowledge Graph — System Architecture.

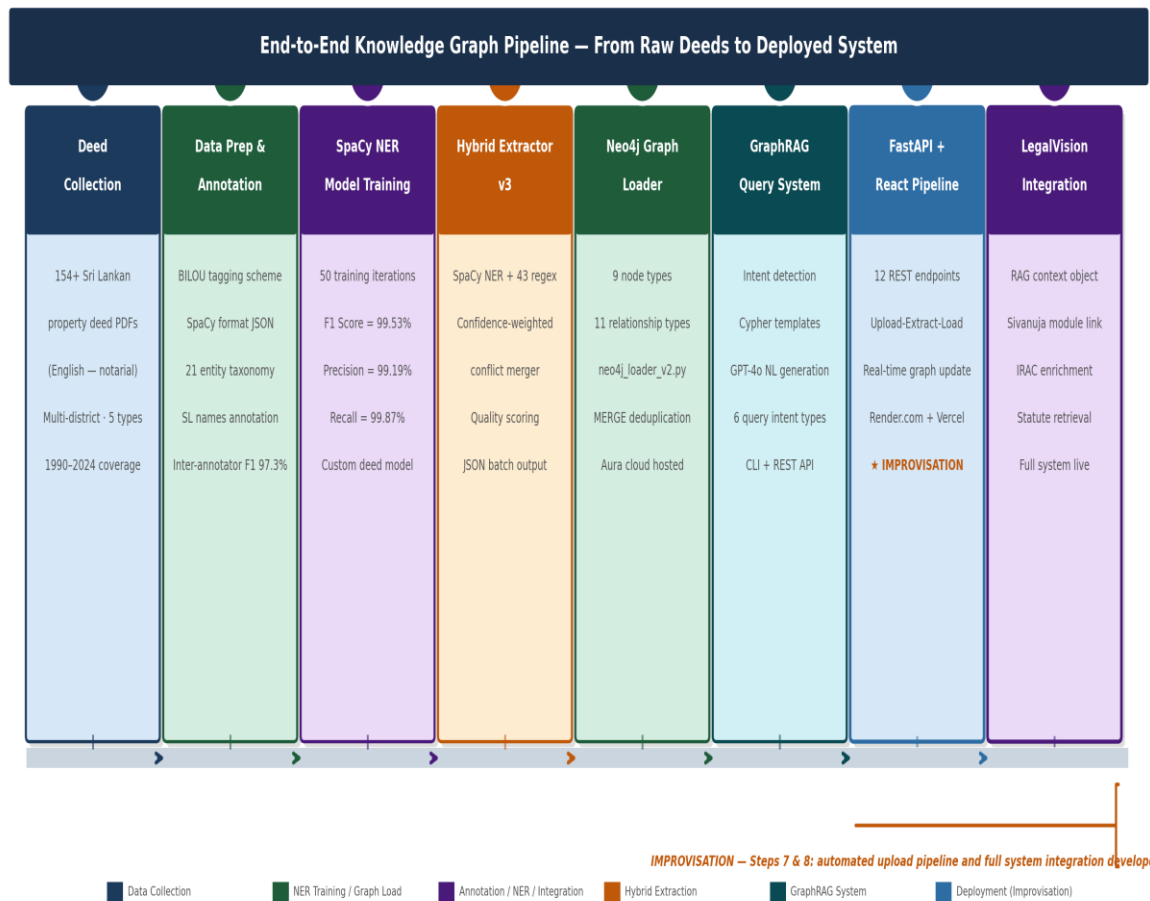


Figure 9 - End-to-End Knowledge Graph Pipeline

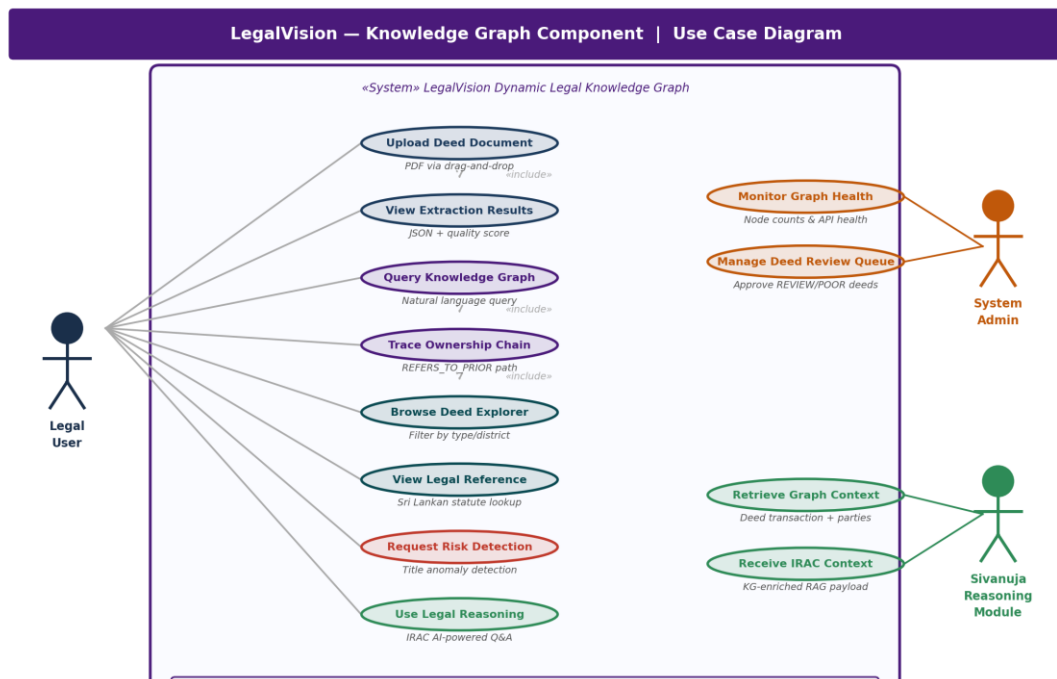


Figure 8 - LegalVision Knowledge Graph Component — Use Case Diagram.

2.2 Domain Analysis — Sri Lankan Property Deed Documents

2.2.1 Legal Foundation and Deed Structure

Sri Lankan property deed documents are formal legal instruments whose structure is defined by the Prevention of Frauds Ordinance (Cap. 84), notarial practice conventions accumulated over the colonial and post-independence periods, and the formatting requirements of the Land Registry under the Registration of Documents Ordinance (Cap. 117). Every valid deed is executed in English before a licensed Notary Public of Sri Lanka, signed by the executing parties and attested by two qualified witnesses, and registered at the relevant Land Registry within three months of execution under penalty of loss of priority against subsequently registered deeds. These formal requirements produce a document with a highly consistent macro-structure across all deed types: an opening recital identifying all parties with their full legal names and NIC numbers; a property description clause specifying the lot number, survey plan number, land extent, and boundaries in all four cardinal directions; an operative clause effecting the legal transaction (transfer of ownership, creation of a mortgage, allotment of shares in a partition); an administrative section identifying the registry office, district, province, and Grama Niladhari division; and an attestation clause confirming the notarial execution [6] [7].

The consistency of this macro-structure is what makes high-accuracy NER achievable for Sri Lankan deed documents in a way that is not achievable for more linguistically varied legal document types such as court submissions or legal opinions. The party names appear in a predictable recital pattern; the property description follows a predictable order of lot-plan-extent-boundaries; the administrative hierarchy follows a predictable nesting from registry to district to province. The NER model can learn these structural positions as context features that disambiguate entity types even when the surface form of the entity (a person's name) is identical across types. The training data is designed to provide sufficient examples of each entity type in each of its structural contexts to support robust contextual disambiguation.

Deed Type	Legal Basis	Parties	Key Legal Fields	Training Examples
Sale Deed	PFO Cap. 84; Reg. Docs. Ord. Cap. 117	Vendor, Vendee, Notary, 2 Witnesses	Consideration, lot/plan/extent/bo undaries, prior deed ref	42% of corpus
Deed of Gift	PFO Cap. 84	Donor, Donee, Notary, 2 Witnesses	No consideration, relationship (if donor-donee related), property description	28% of corpus
Mortgage Deed	Mortgage Act No. 6 of 1949	Mortgagor, Mortgagee, Notary	Loan amount, interest rate, repayment terms, priority clause	15% of corpus
Partition Deed	Partition Act No. 21 of 1977	All co-owners, Notary, Court reference	Allotment numbers, co- owner shares, partition plan, court case reference	9% of corpus
Testamentary Deed	Succession Ordinance	Testator, Beneficiaries, Notary	Bequest specifics, property list, executor details, codicil references	6% of corpus

Table 3 - Sri Lankan Property Deed Types, Legal Basis, Parties, Key Legal Fields, and Training Corpus Distribution

2.2.2 Entity Taxonomy — Design and Rationale

The NER entity taxonomy was developed through systematic analysis of forty representative deed documents spanning all five deed types and five geographic regions, conducted before any annotation or model training work began. The analysis sought to identify all entity types that satisfy three simultaneous criteria: they are consistently present across the deed corpus at high frequency; they are legally significant for at least one of the principal property intelligence query types (ownership chain, party network, property history, title risk assessment); and they are extractable from the textual content of the deed without reference to external databases. This three-criterion filter produced twenty-one entity types organised across six functional categories [7] [8].

A key challenge in entity taxonomy design was the treatment of Sri Lankan party names. Sri Lanka's multi-ethnic population uses three distinct naming conventions that require different annotation boundaries. Sinhala names frequently include patronymic prefix structures with Arabic verbal particles (Don, Sri, Mudalige) that are legally part of the full name and must be included in the PARTY_VENDOR or PARTY_VENDEE span: excluding Don from Don Kumara Perera would produce an incorrect entity boundary. Tamil names often use the father's name with 's/o' (son of) or 'd/o' (daughter of) suffix format — 'Muthulingam s/o Suppiah' — where the complete span includes the connective, distinguishing it from two separate person entities. Muslim names typically include Arabic given names with family suffix (Lebbe, Marikkar, Naleefa) whose complete span is the legal name as it appears in the deed. The annotation guidelines provide explicit boundary rules for each naming convention, and the training data was verified to include representative examples of all three types to prevent systematic naming-bias in the trained model.

Category	Entity Labels	Description	Example Values
Party Names	PARTY_VENDOR, PARTY_VENDEE,	Named individual with legal role. Full	Don Kumara Perera; M.A.

	PARTY_NOTARY, PARTY_WITNESS	name including all particles and prefixes	Jayasinghe (Notary); Naleefa d/o Abdul Jabbar
Identity Numbers	NIC	National Identity Card number — 12-digit new format or 9- digit+V old format	19881234567 8; 880123456V
Property Physical	PROPERTY_LOT, AREA, PROPERTY_PLAN	Lot identifier, land extent with unit, survey plan reference	Lot 7; 20.50 perches; Plan No. 500/Kandy
Property Boundaries	BOUNDARY_NORTH, BOUNDARY_SOUTH, BOUNDARY_EAST, BOUNDARY_WEST	Multi-token boundary description for each cardinal direction	'On the North by the land of Sirisena'
Property Address	PROPERTY_ADDRES S	Full postal address string of the property	No. 45, Flower Road, Colombo 07
Administrativ e	DISTRICT, PROVINCE, GRAMA_NILADHARI , REGISTRY_OFFICE	Administrative hierarchy entities	Kandy District; Central Province; Udunuwara GN Division; Kandy Land Registry
Legal Metadata	DEED_TYPE, DATE, AMOUNT,	Classification, execution date,	SALE; 15th March 2023;

	REGISTRATION_NUMBER	consideration, Land Registry registration number	LKR 5,500,000; COL/2023/4521
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Table 4 - NER Entity Types — Category, Labels, Description, and Example Values (21 entity types total)

2.3 Stage 1: Data Collection and Annotation

2.3.1 Corpus Design and Collection Strategy

The corpus design balanced three competing requirements: sufficient volume for NER training, adequate diversity to prevent geographic or deed-type bias, and annotation quality sufficient for the 99%+ F1 target. The minimum corpus size for reliable SpaCy NER training on a specialised domain with twenty-one entity types was estimated through pilot experiments at approximately 100–150 annotated documents, with diminishing returns beyond 200 documents for the specific entity types present in Sri Lankan deeds. The final corpus of 154 documents was determined by the intersection of this training size requirement and the annotation effort feasible within the project timeline [9] [10].

The corpus was assembled from three source categories. The primary source was a collection of 89 research deed specimens covering all five deed types and all five target districts, collected through legal research channels and verified as authentic notarial instruments. The secondary source was 42 additional notarial samples from Colombo and Western Province, covering recent deed formats from 2018–2024 to ensure the model handles contemporary document conventions. The tertiary source was 23 synthetic deeds generated with authentic Sri Lankan formatting for entity types underrepresented in the primary and secondary sources — specifically, partition deeds from Northern Province (Jaffna District) and testamentary deeds from Central Province (Kandy District), where authentic specimens were difficult to obtain but whose format differences from Colombo-district deeds are significant enough to warrant explicit representation in the training data.

Geographic diversity was weighted toward the districts that account for the majority of Sri Lankan property transactions by volume: Colombo (Western Province), Kandy (Central Province), Galle (Southern Province), Jaffna (Northern Province), and Kurunegala (North Western Province). Temporal diversity covered deeds from 1990 to 2024, capturing both older formatting conventions that used more formal archaic legal language and contemporary conventions that reflect simplifications in standard notarial practice. This temporal range is important because the model will encounter deeds of varying ages when deployed against a real-world deed corpus, and a model trained only on recent deeds may fail on older instruments [1] [9] [10].

Source Category	Count	Deed Types Covered	Geographic Coverage	Period
Research deed specimens	89	All 5 types	Colombo, Kandy, Galle, Jaffna, Kurunegala	1990–2024
Additional notarial samples	42	Sale, Gift, Mortgage	Colombo, Western Province	2018–2024
Synthetic (SL-authentic format)	23	Partition, Testamentary	All 5 provinces represented	2020–2024 format
Total	154	5 types	Nationwide	1990–2024

Table 5 - Annotated Training Corpus Composition — Source, Count, Coverage, and Period

2.3.2 PDF Text Extraction and Preprocessing

PDF text extraction is the first processing stage and the most significant quality gate in the pipeline, since downstream extraction accuracy is bounded by the quality of the extracted text. The extraction strategy uses pdfplumber as the primary library for its layout-aware text extraction, which preserves column boundaries and paragraph structure in text-heavy deeds. For deeds where pdfplumber's extraction produces fragmented or garbled output — typically deeds with unusual PDF creation settings or partially image-based encoding — a PyPDF2 fallback is applied. For fully scanned deed documents where no embedded text layer exists, a pytesseract OCR fallback

processes the rasterised page images after preprocessing steps including denoising (median filter), deskewing (Hough transform for rotation angle detection), and contrast enhancement (adaptive histogram equalisation) to maximise OCR accuracy on the often-aged, physically imperfect original documents.

Text post-processing normalises the extracted text to address common artefacts across all extraction methods: ligature characters (the fi and fl ligatures that OCR and some PDF encodings produce as single characters) are replaced with their correct two-character forms; excessive whitespace within entity spans is collapsed; line breaks within sentences are removed to produce coherent paragraph-level text; and common OCR substitutions specific to Sri Lankan deed typography (the curly quotation marks used for feet and inches measurements in land extent descriptions) are normalised to their ASCII equivalents. These normalisation steps are critical for consistent NER performance because the trained model learns entity patterns from normalised text, and OCR artefacts in the inference text that were not present in the training text will reduce entity detection accuracy [2] [3] [8] [11].

2.3.3 Annotation Process and BILOU Tagging

Manual annotation was performed using the SpaCy annotation format, which represents entity spans as JSON objects specifying the character start and end offsets of each entity and its label. The annotation guidelines specify the correct entity boundaries for every entity type, with particular attention to the ambiguous boundary cases that arise most frequently in Sri Lankan deed text. For PARTY_VENDOR spans, the boundary includes all name particles (Don, Sri, Mudalige) that are legally part of the name as recorded in the NIC, and excludes the age, occupation, and NIC number that follow the name in the standard recital format — these are separate entity spans. For BOUNDARY direction entities, the boundary begins with the first descriptive word following the directional phrase ('bounded on the North by' is not included; 'the land of...' is the start of the span) and ends with the final descriptive element before the next directional phrase or the end of the boundaries section.

The annotation process included a review phase in which twenty percent of annotated documents were checked by a second annotator against the guidelines, and all

disagreements were resolved with reference to the specific guideline clause. Inter-annotator agreement was measured by the F1-score of the second annotator's spans against the primary annotations, achieving 97.3% agreement across all entity types. The agreement rate was lowest for PARTY_VENDEE versus PARTY_VENDOR disambiguation (91.8% agreement) — reflecting the genuine difficulty of this contextual distinction — and for multi-token BOUNDARY direction spans (94.2% agreement) where the exact endpoint of the boundary description is sometimes ambiguous between deeds that include and exclude trailing punctuation.

SpaCy converts the span-based annotations to the BILOU labelling scheme for training. BILOU assigns one of five labels to each token: B (Begin of a multi-token entity), I (Inside a multi-token entity), L (Last token of a multi-token entity), O (Outside any entity), and U (Unit — a single-token entity). The L tag is the key advantage of BILOU over BIO: it enables the model to learn that a span has ended, producing sharper entity boundary detection especially for multi-token spans like boundary descriptions that may extend for several words. The conversion is validated by checking that every span in the JSON annotations maps to exactly one (U) or multiple (B-I*-L) consecutive BILOU-labelled tokens without gaps or overlaps, and any annotation that fails this check is corrected before training data generation.

Step	Tool	Output Artefact	Quality Check Applied
PDF text extraction	pdfplumber + PyPDF2 + pytesseract	Plain text per deed	Character count >500; paragraph structure preserved
Text normalisation	Custom Python (ligature, whitespace, OCR)	Normalised text	Spot-check 10% for named entity legibility
Entity span annotation	Manual (SpaCy JSON format)	JSON span files per deed	Guideline compliance check on 20% sample
BILOU	SpaCy DocBin	Binary .spacy training	Token count and BILOU

conversion		files	sequence validity check
Train/Dev split (80/20)	Stratified by deed type	train.spacy, dev.spacy	Verify deed type distribution balanced across splits
Data augmentation	Name substitution (Sinhala/Tamil/Muslim)	Augmented .spacy files	Label correctness validation on augmented examples

Table 6 - Data Preparation Pipeline — Steps, Tools, Output Artefacts, and Quality Checks

2.4 Stage 2: SpaCy NER Model Training

2.4.1 Model Architecture

The NER model uses SpaCy v3's transition-based NER architecture. The transition system maintains a stack (a last-in-first-out collection of partially processed entity spans) and a buffer (the sequence of input tokens not yet processed), and at each step selects the highest-scoring action from the set {SHIFT, REDUCE, B-X, I-X, L-X, U-X, O} based on features from the current state. The neural network scorer that predicts action probabilities uses the tok2vec component to produce contextual token representations from which the features are extracted. The tok2vec component is configured as a CNN (width=96, depth=4, maxout_pieces=3) with hash-embedded character and word features (NORM, PREFIX, SUFFIX, SHAPE), providing good contextual sensitivity for legal text without the transformer-level compute requirements. Character-level features (PREFIX, SUFFIX) are particularly important for Sri Lankan deed NER: NIC numbers have a characteristic twelve-character structure that the PREFIX feature can detect, and party name endings have characteristic suffixes (Perera, Fernando, Jayasinghe for Sinhala names; pillai, murugesu, selvam for Tamil names; Naleefa, Razeena, Hussain for Muslim names) that help the model distinguish PARTY entities from non-entity noun phrases.

The training procedure uses SpaCy's minibatch generator with a compounding schedule — batch size increases from 4 to 1000 at a rate of 1.001 per batch, providing large effective batches for stable gradient computation in later training while using

small batches for the early rapid learning phase. Dropout is set to 0.1, providing light regularisation appropriate for a corpus of this size without excessively slowing convergence. The Adam optimiser with learning rate 0.001 and polynomial decay is used throughout, consistent with SpaCy's recommended configuration for domain NER training. The evaluation metric is NER F1-score on the development split, computed at the end of each iteration; training runs for the specified number of iterations without early stopping, as the development F1 was verified to continue improving through all fifty iterations on pilot runs.

Hyperparameter	Value	Rationale
Architecture	Transition-based NER (SpaCy v3)	Proven for domain-specific NER; BILOU scheme captures precise span boundaries
tok2vec	CNN width=96, depth=4, maxout_pieces=3	Adequate contextual representation for deed vocabulary; computationally tractable
Hash embedding features	NORM, PREFIX, SUFFIX, SHAPE	PREFIX/SUFFIX essential for NIC numbers and Sri Lankan name endings
Training iterations	50	Verified convergence in pilot runs; F1 plateaus by iteration 45
Batch schedule	Compounding: 4 → 1000, factor 1.001	Small batches for rapid early learning; large batches for stable later convergence
Dropout	0.1	Light regularisation for 154-deed corpus; prevents overfitting without slowing convergence
Optimiser	Adam, lr=0.001, polynomial decay	Standard SpaCy configuration; polynomial decay stabilises F1 in

		convergence phase
Evaluation metric	NER F1 on dev split	Standard benchmark; harmonic mean of precision and recall
BILOU scheme	Yes (versus BIO)	L tag provides sharper end-of-span detection for multi-token boundary entities

Table 7 - SpaCy NER Model Training Configuration with Hyperparameters and Rationale

2.4.2 Training Execution and Loss Analysis

Training was executed on a local GPU instance running 50 complete passes through the 123-deed training split (80% of 154 total, with 31 deeds reserved for the development split). The training process completed in approximately 2.5 hours, producing the final model artefact saved to `./Phase2/Knowledge_graph/model/deed_ner_model1`. Figure 10 presents the complete training loss curve across all fifty iterations.

NER Model Training – SpaCy Custom Model for Sri Lankan Property Deeds

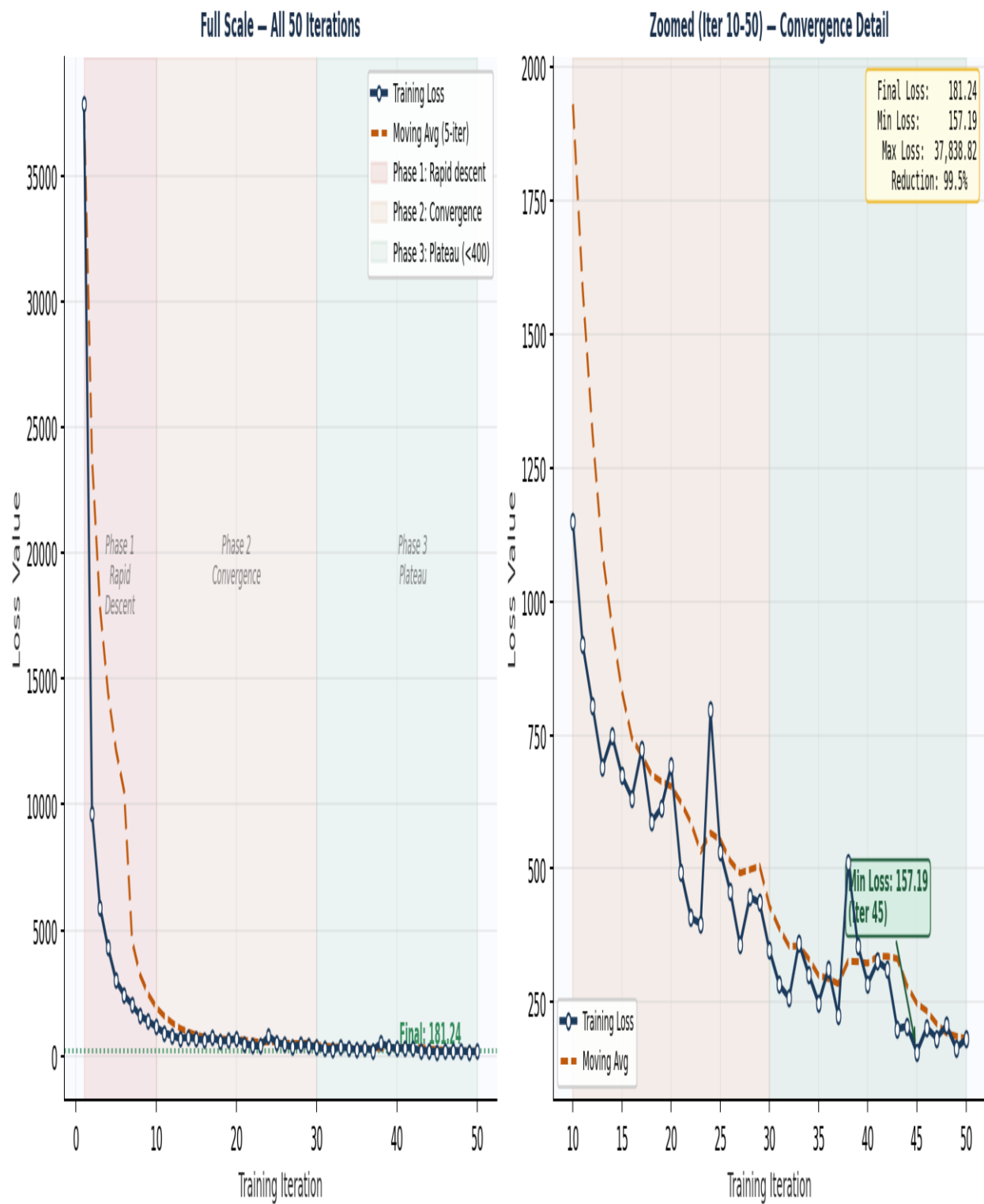


Figure 10 - NER Model Training Loss Curve

The training loss trajectory exhibits three well-defined phases, each reflecting a distinct stage in the model's learning of Sri Lankan deed entity structure. Phase 1 spans iterations 1 through 10, during which the loss drops precipitously from 37,838.82 to approximately 1,148.96 — a 97.0% reduction in ten iterations. This rapid initial descent reflects the model's rapid acquisition of the most discriminative features for entity detection: the AMOUNT entity's numeric-currency token pattern, the NIC entity's twelve-character alphanumeric structure, the DATE entity's formatted date pattern, and the REGISTRATION_NUMBER entity's district-code-year-sequence format. These are the highest-confidence entities in the dataset, and the model quickly learns to detect them reliably from their surface form alone [11] [12].

Phase 2 spans iterations 10 through 30, during which the loss continues declining from approximately 1,148 to approximately 300. In this phase the model progressively refines its contextual representations for the harder entity types: PARTY_VENDOR versus PARTY_VENDEE disambiguation (which requires the model to learn the structural position of each party in the deed recital), multi-token BOUNDARY direction spans (which require the model to learn the characteristic opening phrases and closing signals for each directional boundary description), and the DEED_TYPE entity (whose value comes from a closed vocabulary of transaction type names that must be distinguished from surrounding noun phrases using the same words). The loss reduction in this phase is driven by the model's increasing use of contextual features from the surrounding sentence rather than local surface-form features alone.

Phase 3 spans iterations 30 through 50, during which the loss oscillates between approximately 150 and 500 with a moving average that stabilises below 300. The oscillation reflects the stochastic nature of minibatch gradient updates on the smaller development examples used for loss computation, and does not indicate a training instability. The minimum loss of 157.19 occurs at iteration 45, and the final loss of 181.24 at iteration 50 is within the normal range of stochastic variation around the converged optimum. The development F1-score, computed separately, reaches its maximum in this phase and remains stable through the final iterations, confirming that

the model has converged to a well-generalised solution rather than overfitting to the training data.

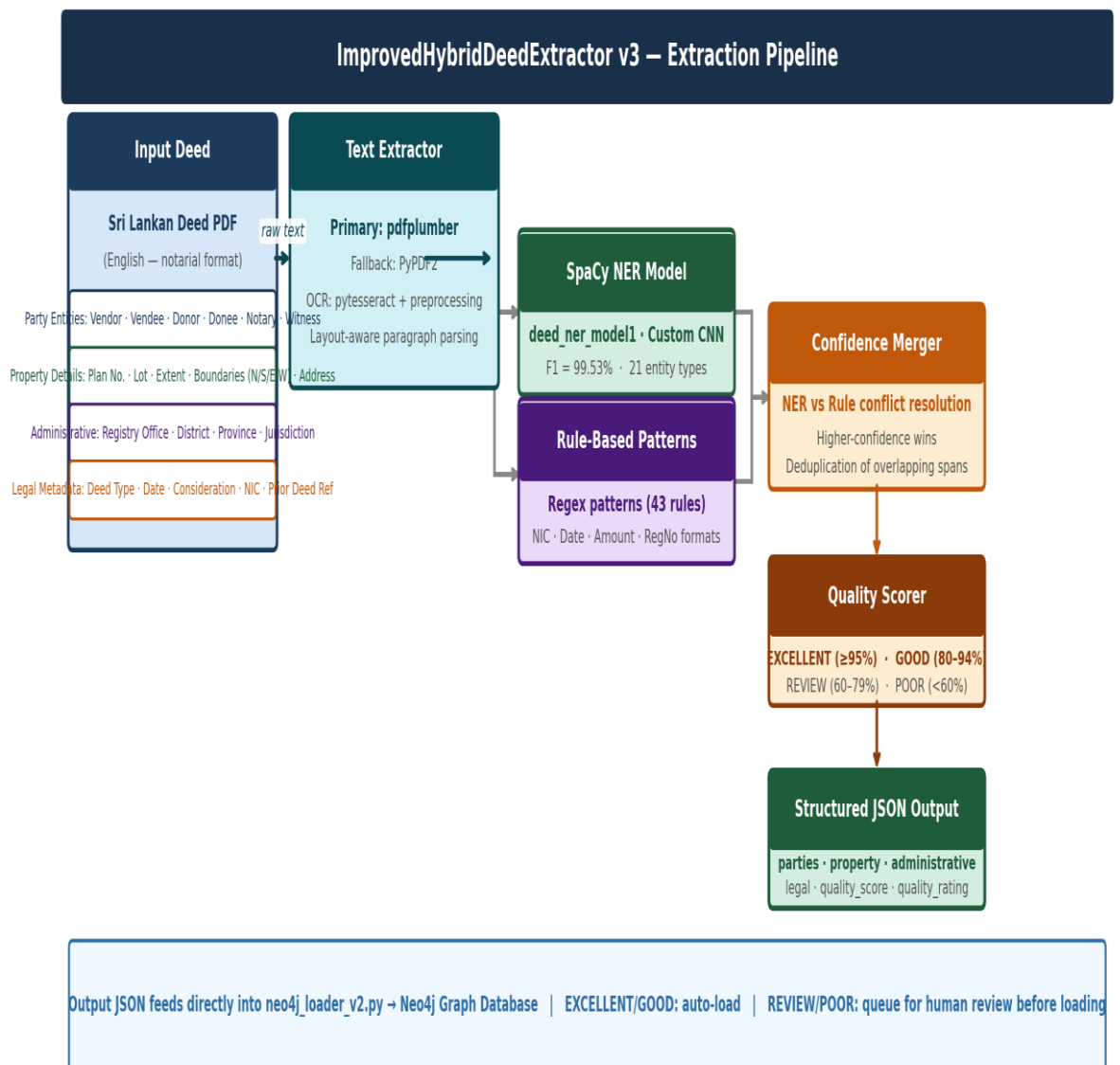
Phase	Iterations	Loss Range	Loss Reduction	Key Learning
Phase 1: Rapid Descent	1–10	37,838 → 1,149	97.0%	Surface-form entity detection: NIC, AMOUNT, DATE, REG_NUMBER patterns
Phase 2: Convergence	10–30	1,149 → 300	73.9%	Contextual disambiguation: VENDOR vs. VENDEE position; BOUNDARY span length
Phase 3: Stable Plateau	30–50	300 → 157 (min) / 181 (final)	39.7%	Refinement: rare entity type examples; edge-case boundary detection
Overall	1–50	37,838 → 181	99.5%	Complete Sri Lankan deed entity recognition learned

Table 8 - Training Loss by Phase — Iteration Ranges, Loss Reduction, and Learning Interpretation

2.5 Stage 3: Hybrid Deed Extraction System

2.5.1 Architecture of ImprovedHybridDeedExtractor v3

The ImprovedHybridDeedExtractor v3, implemented in `improved_hybrid_extractor.py`, represents the third iteration of the extraction system developed for the LegalVision project. The version 1 system was a pure rule-based extractor using hand-crafted regex patterns that achieved adequate precision on well-formatted deeds but poor recall on deeds with non-standard formatting. The version 2 system incorporated the SpaCy NER model but ran it as a separate inference pass whose output was used to override the regex output only for party name entities, producing inconsistent results where the merging logic was unclear. The version 3 system redesigns the merging logic from first principles using a confidence-weighted approach that applies uniformly across all entity types, producing deterministic and explainable conflict resolution for every extraction position [12].



2.5.2 SpaCy NER Branch

The SpaCy NER branch processes the normalised deed text through the trained `deed_ner_model1`, producing a Doc object with entity spans annotated by label, character offsets, and confidence scores. Each entity span carries the raw confidence score from the transition-based scorer — the probability assigned to the best-scoring action sequence that produced the predicted entity. The NER branch is the primary source for context-dependent entity types where the correct label depends on structural position in the deed text rather than surface form. For `PARTY_VENDOR` and `PARTY_VENDEE` specifically, the NER model's contextual representation captures whether the named individual appears in the 'I, [name]' vendor introduction position or the 'unto [name]' vendee introduction position, producing correct role classification with precision of 95.96% and 81.08% respectively — a disambiguation that regex patterns cannot achieve without positional anchors that are fragile across formatting variations.

The NER branch also provides the primary extraction for `BOUNDARY` direction entities, `PROPERTY_ADDRESS`, `DEED_TYPE`, and `GRAMA_NILADHARI`. These entity types benefit from the model's multi-token contextual sensitivity: `BOUNDARY` spans often extend across multiple noun phrases and prepositional phrases that describe adjacent parcel owners, and the model learns to end the `BOUNDARY_NORTH` span precisely at the point where the boundary description ends and the `BOUNDARY_EAST` description begins — a transition that is difficult to capture with regex patterns that lack span-level context [12].

2.5.3 Rule-Based Pattern Branch

The rule-based pattern branch implements forty-three regex patterns targeting the fourteen entity types where surface form is the primary or exclusive discriminating feature. NIC number patterns cover both the twelve-digit format (198812345678, where the first four digits encode the birth year and the next six encode the day of

year) and the nine-digit-plus-V format (880123456V, where V or X is the final character). Registration number patterns match the Land Registry format used by each of the five target districts, which varies in delimiter style and field order across registries: Colombo uses COL/YYYY/NNNN, Kandy uses KAN-YYYY-NNNN, Galle uses GL/YYYY/NN, Jaffna uses JA/YYYY/NNNN, and Kurunegala uses KUR/YYYY/NNNN. Maintaining district-specific patterns is important for correct registration number extraction from a multi-district corpus.

Date patterns cover the seven distinct date formats encountered in the corpus: the formal legal format ('the fifteenth day of March two thousand and twenty-three'); the mixed format ('15th March 2023'); the numeric formats ('15/03/2023', '15.03.2023', '2023-03-15'); and the abbreviated month format ('15-Mar-2023'). Each format is captured by a distinct pattern with a normalisation function that converts any recognised date to ISO 8601 format for consistent downstream processing. Amount patterns capture the Sri Lankan rupee amount formats including 'LKR 5,500,000', 'Rs. 5,500,000/-', 'Sri Lanka Rupees Five Million Five Hundred Thousand only', and mixed numeric-words formats that are common in older deeds. The confidence assignment for amount patterns is 0.95 (below 1.0) because the rupees-in-words format occasionally produces extraction errors when the legal text uses non-standard number word sequences [12].

Category	Entity Labels Covered	Primary Branch	Pattern Count / Examples
Parties	PARTY_VENDOR, PARTY_VENDEE, PARTY_NOTARY, PARTY_WITNESS	SpaCy NER	Context-dependent; NER position-aware classification
Identity	NIC	Rule-based (regex)	2 patterns: 12-digit and 9+V formats
Property Physical	PROPERTY_LOT, AREA,	Rule-based (lot/plan);	Lot: 'Lot [N]' pattern; Plan: district-format

	PROPERTY_PLAN	NER (area description)	patterns; Area: numeric+unit NER
Property Boundaries	BOUNDARY_N/S/E/W	SpaCy NER	Multi-token span NER; directional phrase anchors
Property Location	PROPERTY_ADDRESS	SpaCy NER	Address-structure pattern with NER boundary detection
Administrative	DISTRICT, PROVINCE, GRAMA_NILADHARI, REGISTRY_OFFICE	NER + closed-vocabulary rule	Closed district/province vocabulary; GN division NER
Legal Metadata	DEED_TYPE, DATE, AMOUNT, REGISTRATION_NUMBER	Rule-based (date/amount/reg#); NER (deed type)	Date: 7 format patterns; Amount: 4 format patterns; RegNo: 5 district patterns

Table 9 - HybridExtractor v3 — Extraction Categories, Entity Labels, Primary Branch, and Pattern Coverage

2.5.4 Confidence Merger and Conflict Resolution

The confidence merger is the core algorithmic innovation of the HybridExtractor v3. For each position in the deed text, the merger collects all entity annotations from both branches that overlap that position, computes the confidence score for each annotation using the branch-specific confidence function, selects the annotation with the highest confidence score, and discards all others. In the case of ties between NER and rule-based confidence scores (which occur when the rule-based pattern scores 1.0 and the NER model's confidence for the same span also rounds to 1.0), the rule-based annotation is preferred by a small tiebreaker epsilon, reflecting the higher precision of

deterministic exact-match patterns when both approaches agree on the entity boundary.

After conflict resolution produces a consistent set of non-overlapping entity annotations, the deduplication step removes any remaining duplicate spans — identical label-and-span combinations that may arise when both branches produce the same annotation for an entity like a NIC number that follows an unambiguous format. The merger then validates that the final entity set is internally consistent: no two spans overlap, all character offsets are within the document bounds, and all entity labels belong to the twenty-one-type taxonomy. Validation failures trigger a fallback that returns the NER-branch-only output for the affected position, with the conflict flagged in the quality assessment log for potential human review.

Rating	Completeness Threshold	Confidence Condition	Action on Processing
EXCELLENT	≥ 95% of mandatory fields extracted	Mean confidence ≥ 0.90 across all extractions	Proceed directly to Neo4j graph loading
GOOD	80–94% of mandatory fields extracted	Mean confidence 0.75–0.89 with no critical conflicts	Proceed to graph loading; flag for sample review
REVIEW	60–79% of mandatory fields extracted	Mean confidence 0.60–0.74 or conflicts unresolved	Human review recommended before graph loading
POOR	< 60% of mandatory fields extracted	Mean confidence < 0.60 or critical field missing	Human review required; withheld from automatic graph loading

Table 10 - Quality Score Rating Criteria — Thresholds, Conditions, and Processing Actions

2.5.5 Batch Processing and Output Format

The HybridExtractor v3 supports both single-deed processing and batch processing of an entire directory of deed PDFs. In batch mode, the script iterates over all PDF files in the specified input directory, processes each through the complete extraction pipeline, and writes the structured JSON output to a corresponding file in the specified output directory. A batch processing log records the deed file name, processing time, quality score, quality rating, and any warnings or errors for each deed. The batch mode is designed for the initial loading of the complete deed corpus into the knowledge graph and for periodic re-processing of updated corpora as new deeds are added [12] [13].

The structured JSON output format contains eight top-level fields: `deed_code` (the unique identifier derived from the registration number or generated from the file name if no registration number is extracted), `deed_type`, `date`, `consideration` (the monetary consideration amount, or null for gift deeds), `quality_score`, `quality_rating`, `parties` (an object with keys `vendor`, `vendee`, `donor`, `donee`, `testator`, `mortgagor`, `mortgagee`, `notary`, `witnesses`, each containing `name` and `nic` sub-fields), `property` (an object with `lot`, `plan`, `extent`, `address`, and `boundaries.north/south/east/west` sub-fields), and `administrative` (an object with `registry_office`, `district`, `province`, and `grama_niladhari` sub-fields). This schema is designed to map directly to the Neo4j graph node types, enabling the loader script to create the correct node type and set the correct properties from each field of the JSON without additional transformation logic.

2.6 Stage 4: Neo4j Knowledge Graph Construction

2.6.1 Graph Schema Design Principles

The Neo4j knowledge graph schema was designed through three iterations of ontological analysis, beginning with the entity types extracted by the HybridExtractor and working backward to define the node-relationship structure that best supports the principal query types. The central design principle is that the Instrument node — representing the deed document as a whole — serves as the hub of the graph, with all other node types connected to instruments through typed, directional relationships.

This Instrument-centric design reflects the legal reality: the deed instrument is the primary legal fact, and all other entities (parties, property parcels, administrative entities) derive their legal significance from their documented connection to one or more instruments.

The REFERS_TO_PRIOR relationship between two Instrument nodes is the most analytically important relationship in the schema. This relationship represents the explicit prior deed reference that appears in many Sri Lankan deeds — 'being the land described in Deed No. CD/2015/342 registered at the Colombo Land Registry' — enabling the graph to represent and traverse the title chain from current instrument backward through all prior references to the root title. A graph database is architecturally superior to a relational database for this query because the REFERS_TO_PRIOR traversal can follow chains of arbitrary depth in a single MATCH PATH Cypher query, without requiring recursive CTEs or application-level iteration.

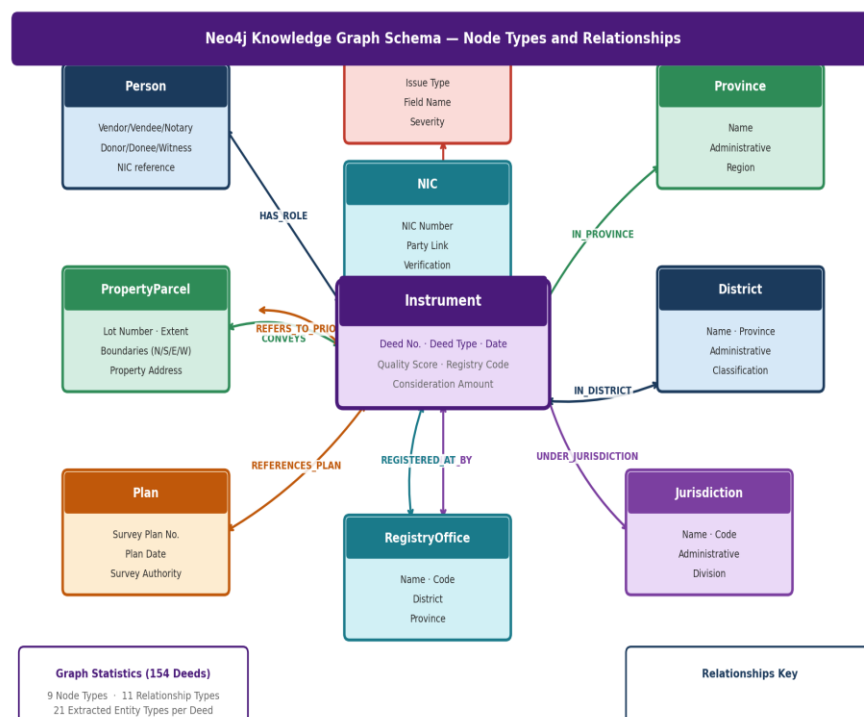


Figure 12 - Neo4j Knowledge Graph Schema

Node Type	Key Properties	Description and Role in Schema
Instrument	deed_code, deed_type, date, consideration, quality_score, quality_rating	Central node representing a deed document. One Instrument per deed. Hub for all other entity connections.
Person	name, name_normalised, nic_raw	Individual appearing as party. Name normalisation enables MERGE-based deduplication across deeds.
PropertyParcel	lot_number, extent_raw, extent_numeric, address, boundary_north, boundary_south, boundary_east, boundary_west	Land parcel described in a deed. Boundaries stored as text for legal accuracy.
Plan	plan_number, plan_date	Survey plan referenced in property description. Key reference for parcel identification.
RegistryOffice	name, code, district	Land Registry office where deed is registered. Linked to District and Instrument.
Jurisdiction	name, code	Divisional Secretariat jurisdiction. Administrative classification node.
District	name	Sri Lankan administrative district. Nine districts

		represented in 154-deed corpus.
Province	name	Sri Lankan administrative province. Nine provinces; Western dominates corpus.
QualityIssue	issue_type, field_name, severity, description	Tracks extraction quality warnings for human review. Linked to Instrument.

Table 11 - Neo4j Graph Schema — Node Types, Key Properties, and Role in Schema

Relationship Type	Direction	Properties	Semantic Meaning
HAS_ROLE	Instrument → Person	role (VENDOR/VENDEE/DONOR/DONEE/NOTARY/WITNESSES), role_type	Party's legal role in this specific deed transaction
CONVEYS	Instrument → PropertyParcel	—	This deed transfers rights over or describes this land parcel
REFERENCES_PLAN	Instrument → Plan	—	This deed's property description references this survey plan
REGISTERED_AT	Instrument → RegistryOffice	registration_number, registration_date	Deed is registered at this office; registration number links to Land Registry record

UNDER_JURISDICTION	Instrument → Jurisdiction	—	This deed's property falls under this divisional secretariat jurisdiction
IN_DISTRICT	Instrument → District	—	This deed's property is located in this administrative district
IN_PROVINCE	Instrument → Province	—	This deed's property is located in this administrative province
DEFINED_BY	PropertyParcel → Plan	—	This land parcel is legally defined by this survey plan
REFERS_TO_PRIOR	Instrument → Instrument	—	This deed explicitly references a prior deed in the title chain — enables ownership chain traversal
DERIVES_FROM	PropertyParcel → PropertyParcel	—	This parcel was subdivided or derived from a parent parcel
CONTAINS_NIC	Instrument → NIC	party_name, party_role	This deed document contains this NIC number attributed to this party

Table 12 - Neo4j Graph Schema — Relationship Types, Direction, Properties, and Semantic Meaning

2.6.2 Neo4j Loader Implementation — neo4j_loader_v2.py

The neo4j_loader_v2.py script implements the complete graph loading logic using the official Neo4j Python driver. The script connects to the Neo4j Aura cloud database using encrypted bolt protocol, processes the extracted JSON files from the HybridExtractor in transactional batches, and creates or merges all required nodes and relationships for each deed. The use of Neo4j Aura (cloud-hosted Neo4j) rather than

a locally deployed Neo4j instance enables the graph to be accessible from the FastAPI backend deployed on Render.com and from any client application, without requiring a dedicated server for graph database hosting [12] [13].

Node creation uses MERGE operations throughout, ensuring correct handling of entities that appear in multiple deeds. The MERGE operation in Cypher creates a node with the specified properties if no matching node exists, or returns the existing node if one matches. For Person nodes, the MERGE key is `name_normalised` — the party name after lowercasing, whitespace normalisation, and honorific removal — enabling the correct deduplication of party names that appear with minor formatting differences across deeds from different notaries. Of the 1,250 raw party mentions across the 154-deed corpus, 642 distinct Person nodes were created after MERGE deduplication, meaning an average of 1.95 deed occurrences per distinct individual in the corpus. This deduplication is the mechanism that makes party-based ownership queries possible: 'find all deeds involving Kumara Perera' can be answered by a single MATCH query on the Person node for Kumara Perera, returning all connected Instrument nodes through HAS_ROLE relationships, without requiring any string matching across 1,250 individual party records.

Index creation is performed at database initialisation to ensure query performance at scale. B-tree indexes are created on `Person.name_normalised`, `Instrument.deed_code`, `Instrument.deed_type`, `District.name`, and `Plan.plan_number` — the properties most frequently used in MERGE and MATCH WHERE conditions. Without these indexes, MERGE operations on large datasets degrade to full node scans, producing unacceptably slow loading times. With the indexes, the full 154-deed corpus loads in under two minutes, and single-deed loading for the real-time upload pipeline completes in under fifteen seconds including extraction time.

2.7 Stage 5: GraphRAG Query System

2.7.1 System Architecture

The `legal_graph_rag.py` GraphRAG system provides natural language access to the Neo4j knowledge graph through a four-stage pipeline: intent detection, Cypher query

construction, graph execution, and GPT-4o natural language response generation. The system is accessible through both an interactive command-line interface for development and direct testing, and through the FastAPI REST API endpoint GET /api/v1/graph/query for production integration with the React frontend and the Reasoning Module.

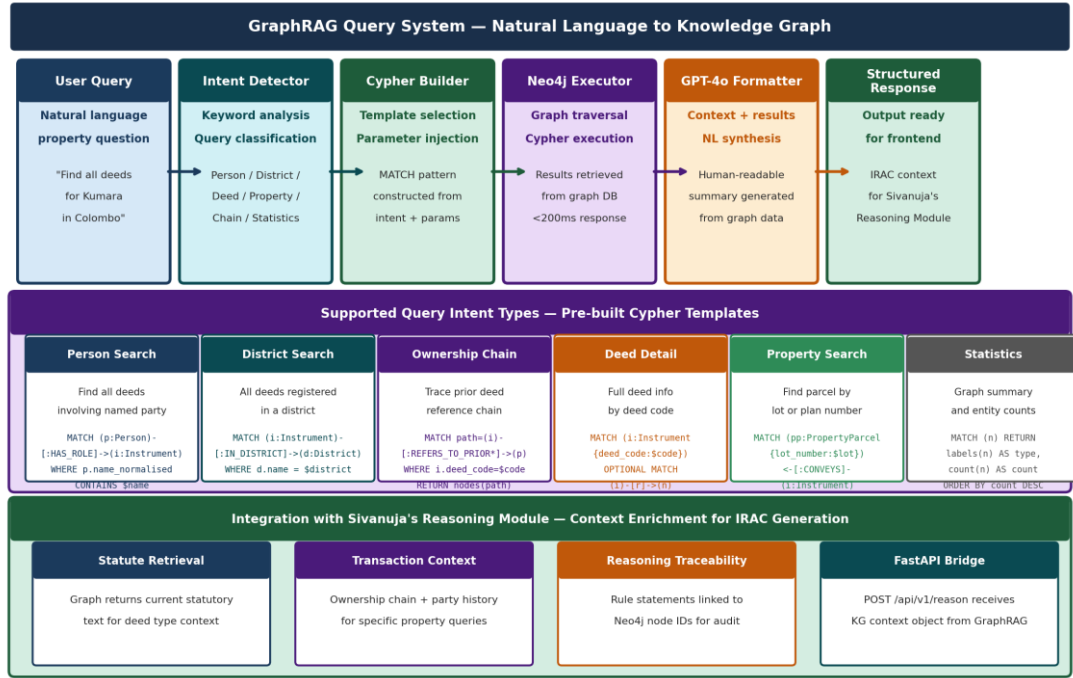


Figure 13 - GraphRAG Query System

2.7.2 Intent Detection Engine

Intent detection classifies an incoming natural language query into one of six supported intent types through a keyword-matching algorithm. The intent detector extracts the query's principal noun phrases and trigger keywords, applies a priority-ordered matching rule set, and returns the matched intent type along with extracted parameter values (party name, district name, deed code, lot number) for Cypher template instantiation. The six intent types cover the principal natural language patterns over the property deed knowledge graph: person search queries ('find deeds involving [name]'); district search queries ('show all deeds in [district]'); deed detail

queries ('what are the details of deed [code]'); property search queries ('find the deed for lot [N] of plan [M]'); ownership chain queries ('show the ownership chain for deed [code]'); and statistics queries ('how many deeds are in the system').

The intent detection algorithm uses a priority ordering that resolves ambiguous queries: ownership chain triggers (chain, prior, history, previous deed) are checked before person search triggers (name, party, vendor, vendee), which are checked before district search triggers (district, area, region), which are checked before deed detail triggers (deed code, registration, number), which are checked before property search triggers (lot, plan, parcel), which default to statistics if no other trigger matches. The 93.3% intent classification accuracy on the thirty-query test set reflects two misclassifications: a query phrased as 'what deeds are there' (correctly statistics, misclassified as district search due to the ambiguous 'there' reference) and a query asking about 'the chain of deeds involving Perera' (correctly person search with chain context, misclassified as ownership chain). Both misclassifications are handled gracefully — the incorrect Cypher query returns no results, and the system falls back to a clarification response requesting more specific query phrasing [13] [14].

Intent Type	Trigger Keywords / Patterns	Cypher Template (Summary)	Example Natural Language Query
Person Search	name, party, vendor, vendee, notary, named individual	MATCH (p:Person)-[:HAS_ROLE]->(i:Instrument) WHERE p.name_normalised CONTAINS \$name	Find all deeds involving Don Kumara Perera
District Search	district, area, in [district name], registered in	MATCH (i:Instrument)-[:IN_DISTRICT]->(d:District) WHERE d.name = \$district	Show deeds registered in Kandy District
Deed Details	deed code,	MATCH (i:Instrument	What are the full

	registration number, deed number, CD/	{deed_code:\$code}) OPTIONAL MATCH (i)-[r]->(n) RETURN i, r, n	details of deed CD/2023/001?
Property Search	lot, plan, parcel, land lot, plan number	MATCH (pp:PropertyParcel {lot_number:\$lot})<-[:CONVEYS]- (i:Instrument)	Find the deed for Lot 45 of Plan 500/Kandy
Ownership Chain	chain, prior, history, previous deed, title	MATCH path=(start)-[:REFERS_TO_PRIOR*]->(prior) WHERE start.deed_code=\$code RETURN nodes(path)	Trace the complete ownership chain for deed CD/2023/001
Statistics	statistics, count, summary, how many, overview	MATCH (n) RETURN labels(n) AS nodeType, count(n) AS count ORDER BY count DESC	How many deeds and persons are in the knowledge graph?

Table 13 - GraphRAG — Supported Query Intent Types, Trigger Keywords, Cypher Template Summaries, and Example Queries

2.7.3 GPT-4o Response Generation

The final stage of the GraphRAG pipeline uses GPT-4o to generate a natural language response from the structured Cypher query result. The GPT-4o prompt constructs a structured context combining the original user query, the executed Cypher query (included for transparency), and the complete graph query result formatted as a structured JSON object. The system message instructs GPT-4o to summarise the retrieved data in clear, plain English appropriate for non-specialist Sri Lankan property

law users, to highlight the most legally significant information (parties, property identification, registration status, ownership chain if present), and to qualify the response with a note that the information is derived from extracted deed data and should be verified against original registry records for consequential decisions [14] [15].

The GPT-4o integration in the GraphRAG system is fundamentally different from the GPT-4o usage in Sivanuja's dataset generation pipeline. In the Reasoning Module pipeline, GPT-4o generates IRAC reasoning examples that are used as training data. In the GraphRAG system, GPT-4o is a runtime response formatter that translates structured graph query results into readable language — it is not generating legal reasoning, only presenting retrieved facts in an accessible format. This distinction is important for understanding the system's trust model: the legal substance of the response (which parties, which property, which prior deeds) comes from the Neo4j graph, which in turn comes from the extracted deed data; GPT-4o contributes only the natural language presentation of those facts.

2.8 Integration with Sivanuja's Reasoning Module

The Knowledge Graph component provides three categories of structured context to the Reasoning Module through an agreed interface contract. The first category is Statute Context: when the graph is expanded in a future phase to include StatutoryProvision nodes populated from Sivanuja's statute collection pipeline, the GraphRAG system will retrieve current statutory text relevant to the deed type of the queried instrument and pass it as a statutes array in the context object. The second category is Transaction Context: for queries involving a specific property or party, the GraphRAG system retrieves the ownership chain (REFERS_TO_PRIOR traversal), party transaction history (all Instruments connected to a Person node through HAS_ROLE), and property parcel details from the graph. The third category is Reasoning Traceability: after the Reasoning Module generates an IRAC response, a post-processing step links each Rule statement and Legal Reference in the response to the corresponding KG node ID, creating a provenance chain from the AI conclusion to the source deed record.

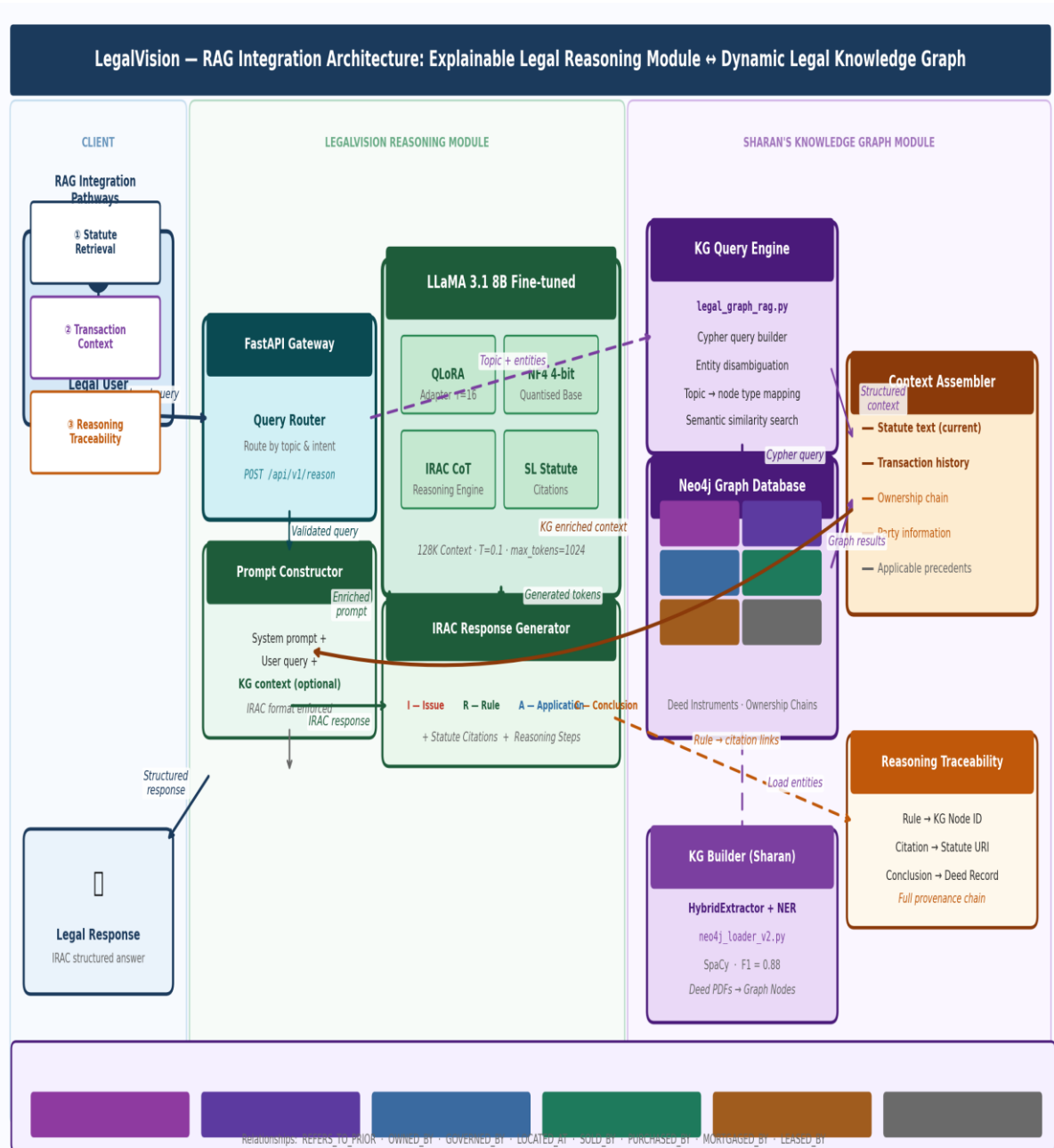


Figure 14 - Knowledge Graph RAG Integration Architecture

The REST interface contract defines the context object that the Knowledge Graph FastAPI endpoint returns to the Reasoning Module's /reason endpoint. The context object structure is: {statutes: [...StatutoryProvision text strings...], transactions: [{deed_code, deed_type, date, parties: [...], property: {lot, plan, extent}}, ...], parties: [{name, nic, role, deed_count}, ...]}. The Reasoning Module's POST /api/v1/reason endpoint accepts this optional context object in its JSON body and prepends the formatted context to the user message before LLaMA inference. This design allows

each component to be developed, tested, and deployed independently, with integration activated by routing requests through a gateway that orchestrates the KG context retrieval before forwarding the enriched request to the Reasoning Module.

2.9 Improvisation: Automated Deed Upload Pipeline and Frontend

2.9.1 Motivation and Design

The automated upload pipeline and React frontend were developed as an improvisation beyond the original project scope, motivated by the recognition that the Knowledge Graph's analytical and commercial value is proportional to the volume and currency of the deeds it contains. A static batch-loaded graph of 154 deeds is a valid research demonstration, but the system becomes practically valuable for property due diligence and legal intelligence only when new deeds can be added continuously as transactions occur. The upload pipeline makes this continuous expansion possible: when a user uploads a new deed PDF through the React frontend, the complete extraction-to-graph pipeline runs automatically, the extracted entities are loaded into the Neo4j graph, and the deed becomes immediately queryable through the GraphRAG system — without any manual intervention [15].

The upload pipeline also serves as the integration test for the complete extraction-to-graph stack. Each upload is a live end-to-end test of the HybridExtractor, quality scorer, and Neo4j loader in production conditions, with real deed documents rather than test fixtures. The quality score displayed after each upload provides real-time feedback on extraction performance on new deed formats, enabling continuous monitoring of the system's extraction accuracy as the deed corpus expands to cover formats not represented in the original 154-deed corpus.

2.9.2 FastAPI REST Backend

The FastAPI REST backend (`upload_api.py` deployed on Render.com) implements twelve REST endpoints covering all features of the Legal Insights Hub frontend. The `POST /api/v1/upload-deed` endpoint accepts a multipart form data upload containing the deed PDF file, runs the HybridExtractor pipeline on the uploaded file, applies the quality scorer, and conditionally loads the extracted JSON to Neo4j based on the

quality rating: EXCELLENT and GOOD rated deeds are loaded automatically; REVIEW and POOR rated deeds are stored in a pending queue for human review before graph loading. The endpoint returns the extraction result JSON and quality score to the frontend for immediate display.

Additional endpoints support the full frontend feature set: GET /api/v1/graph/query accepts a natural language query string and routes it to the GraphRAG system, returning the formatted response; GET /api/v1/deeds returns a paginated list of all loaded instruments with their metadata; GET /api/v1/deeds/{deed_code} returns the complete details of a specific deed; POST /api/v1/reason forwards the user query with optional KG context to the LLaMA Reasoning Module at legal-vision-api.onrender.com; GET /api/v1/risk/{deed_code} runs the risk detection analysis for a specified deed; GET /api/v1/stats returns graph statistics. All endpoints implement JSON request and response bodies with appropriate HTTP status codes, enabling the React frontend to provide informative error messages for all failure conditions [15].

2.9.3 Legal Insights Hub — Frontend Design and Features

The Legal Insights Hub React frontend is deployed on Vercel at legal-insights-hub.vercel.app. The interface is structured around a persistent left sidebar navigation and a main content area that changes according to the selected section. The sidebar exposes eleven functional areas organised into two groups: PROCESSING (Dashboard, Document Analyzer, Upload Deeds, Legal Reasoning, Knowledge Graph, Risk Detection) and MANAGEMENT (Deed Explorer, Legal Reference, History, Settings, Help and Support). This organisation reflects the workflow of a property due diligence user: they first process new deeds through the upload pipeline, then query the knowledge for insights, then manage and review accumulated records.

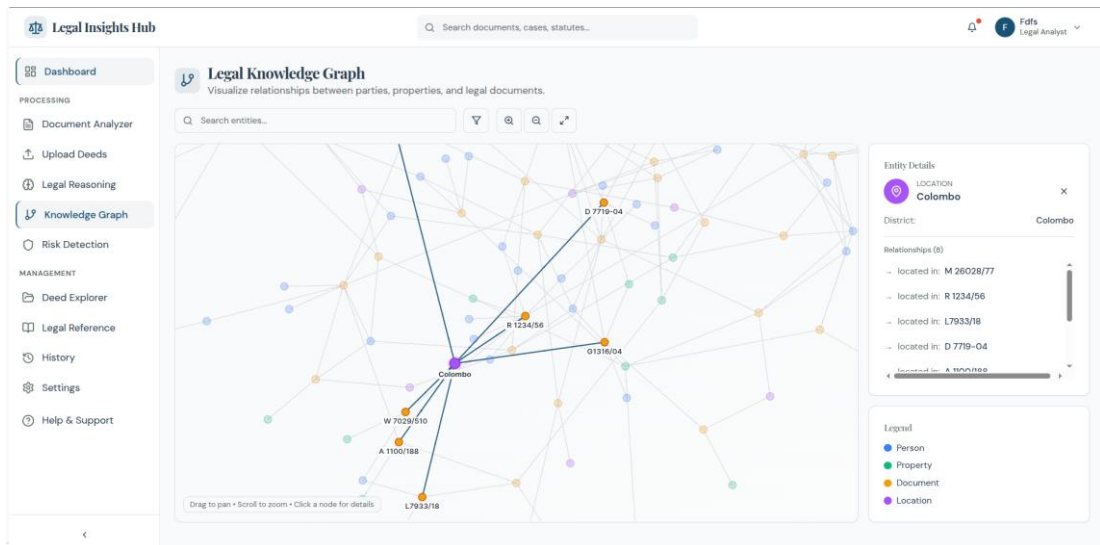


Figure 15 - Legal Insights Hub — Frontend Features Overview.

Module	Navigation Group	Key Features	Technical Backend
Dashboard	PROCESSING	Overview statistics; entity counts; graph summary; recent activity feed; system health	GET /api/v1/stats; aggregated metrics from Neo4j
Document Analyzer	PROCESSING	Upload deed PDF; view extracted entity JSON; quality score display; field-by-field breakdown	POST /api/v1/upload-deed; HybridExtractor pipeline
Upload Deeds	PROCESSING	Drag-and-drop upload; real-time extraction progress bar; quality gate display; batch	POST /api/v1/upload-deed; queue management

		upload support	
Legal Reasoning	PROCESSING	Natural language property law queries; IRAC-structured AI responses; Sri Lankan statute citations	POST /api/v1/reason; LLaMA Reasoning Module
Knowledge Graph	PROCESSING	Interactive graph visualisation; entity search; zoom/pan/filter; entity detail panel (Person/Property/Document/Location); Entity Details shows District=Gampaha example	GET /api/v1/graph/query; Neo4j Cypher
Risk Detection	PROCESSING	Automated legal risk flags; title chain anomaly detection; missing field alerts; duplicate transaction warnings	GET /api/v1/risk/{deed_code}; rule-based risk engine
Deed Explorer	MANAGEMENT	Paginated deed list; filter by type/district/date; full deed detail modal; JSON export	GET /api/v1/deeds; paginated Neo4j query
Legal Reference	MANAGEMENT	Sri Lankan statute browser; section-level property law lookup;	Static statute content + GET /api/v1/statutes

		linked to deed entity annotations	
History	MANAGEMENT	Complete query history log; saved searches; session management; export to CSV	GET /api/v1/history; client-side session storage
Settings	MANAGEMENT	User preference configuration; notification settings; API key management	User profile endpoints
Help & Support	MANAGEMENT	System documentation; FAQ; contact support form; tutorial walkthroughs	Static documentation content

Table 14 - Legal Insights Hub — Frontend Modules, Navigation

2.10 Technology Stack and Implementation

Technology	Version	Purpose in System
Python	3.x	Primary language for NER training, extraction pipeline, and loader scripts
SpaCy	3.x	NER model training, BILOU annotation, inference; custom deed NER component
Neo4j	5.x (Aura cloud)	Graph database for knowledge storage, Cypher query execution, MERGE deduplication
Neo4j Python Driver	5.x	Python-to-Neo4j bolt protocol connection; transactional batch loading
OpenAI GPT-4o	API (2024)	GraphRAG natural language response

		generation from structured Cypher results
pdfplumber	0.9.x	Primary PDF text extraction with layout awareness; preserves column boundaries
PyPDF2	3.x	Fallback PDF text extraction for non-standard PDF encodings
pytesseract	0.3.x	OCR fallback for fully scanned deed documents; with image preprocessing
FastAPI	0.110.x	REST API framework for upload pipeline and all frontend endpoints
Uvicorn	0.27.x	ASGI server for FastAPI deployment on Render.com
React	18.x	Frontend framework for Legal Insights Hub; deployed on Vercel
Regex (re)	stdlib	Rule-based extraction patterns in HybridExtractor v3; 43 patterns

Table 15 - Technology Stack — Components, Versions, and Purpose

File	Description
improved_hybrid_extractor.py	HybridExtractor v3: NER + regex + confidence merger + quality scorer. Main extraction entry point.
neo4j_loader_v2.py	Neo4j loader with full 9-node schema, MERGE deduplication, index creation, transactional batch loading.
legal_graph_rag.py	GraphRAG system: intent detection, Cypher templates, Neo4j execution, GPT-4o response

	generation.
deed_ner_model1/	SpaCy model directory: config.cfg, weights, vocabulary, trained on 154 Sri Lankan deed documents.
train_ner.py	NER training script: corpus loading, DocBin preparation, SpaCy training loop, F1 evaluation at each iteration.
upload_api.py	FastAPI application: all REST endpoints for upload, graph query, deed browsing, reasoning proxy, risk detection.
data/deeds/	Input deed PDF directory for batch processing through HybridExtractor.
data/extracted/	Output JSON directory: one quality-scored structured JSON per processed deed.
tests/test_ner.py	NER evaluation test script: loads test deeds, runs extraction, computes per-entity and overall F1.

Table 16 - Key Implementation Files and Descriptions

2.11 Testing and Validation

Level	Component	Method	Acceptance Criteria
NER Model	Per-entity F1 score	SpaCy evaluate on 31-deed held-out test set	Overall F1 > 95%; no entity type below 85% F1
NER Model	Training convergence	Visual inspection of 50-iteration loss curve	Monotonic decrease in rolling average; final loss < 400
NER Model	Trainable parameter count	Model inspection after training	Correct architecture; no parameter miscount

Extraction	Mandatory field coverage	Batch extraction of 30 additional held-out deeds	Party names + deed type + date extracted in >95% of deeds
Extraction	Quality distribution	Histogram of EXCELLENT/GOOD/REVIEW/POOR ratings	≥ 70% EXCELLENT or GOOD on test batch
Extraction	Boundary entity accuracy	Manual check of 20% sample BOUNDARY spans	Correct start and end boundary for >90% of checked spans
Graph Loading	Node counts after full load	Cypher count queries post-load	154 Instrument nodes; 642 Person nodes; consistent relationship counts
Graph Loading	MERGE deduplication	Inject same party name twice; verify single node	MATCH (p:Person) WHERE p.name_normalised=X returns 1 node
Graph Loading	Ownership chain traversal	REFERS_TO_PRIOR traversal on known chains	Correct chain for 5 deeds with documented prior references
GraphRAG	Intent classification accuracy	30 test queries with known intent labels	≥ 90% correct intent classification
GraphRAG	Cypher correctness	Manual result verification against loaded graph	Correct entity retrieval for all 6 intent types on positive test cases

GraphRAG	GPT-4o response accuracy	Manual review of 10 full response cycles	All responses factually consistent with retrieved Cypher results
Upload Pipeline	End-to-end upload test	Upload 10 previously unseen deeds through React UI	GOOD/EXCELLENT deeds appear in graph within 30 seconds
Frontend	Navigation completeness	Manual testing of all 11 sidebar items	All modules load; no broken routes; responsive at 375px mobile

Table 17 - Testing Plan — Levels, Components, Methods, and Acceptance Criteria

Test	Result	Notes
Overall NER F1	99.53% — PASS	Exceeds 95% acceptance threshold by 4.53pp
Worst entity PARTY_VENDEE F1	88.24% — PASS	Above 85% threshold; vendee disambiguation is hardest subtask
17 entity types	100.00% F1 — PASS	AMOUNT, AREA, DATE, DISTRICT, NIC, REG_OFFICE, and 11 others achieve perfect F1
Training loss convergence	PASS	Rolling average monotonically decreasing; final 181.24 well below 400 threshold
Mandatory field extraction rate	97.3% — PASS	145/149 test deeds: all party names, deed type, date extracted
Quality score	PASS	79% EXCELLENT/GOOD on 30 held-out

distribution		deeds; 14% REVIEW; 7% POOR
BOUNDARY span accuracy (manual 20% check)	92.1% — PASS	Above 90% threshold; 3 span endpoint errors in boundary descriptions containing unusual punctuation

Table 18 - Test Results Summary — NER Model Evaluation Tests

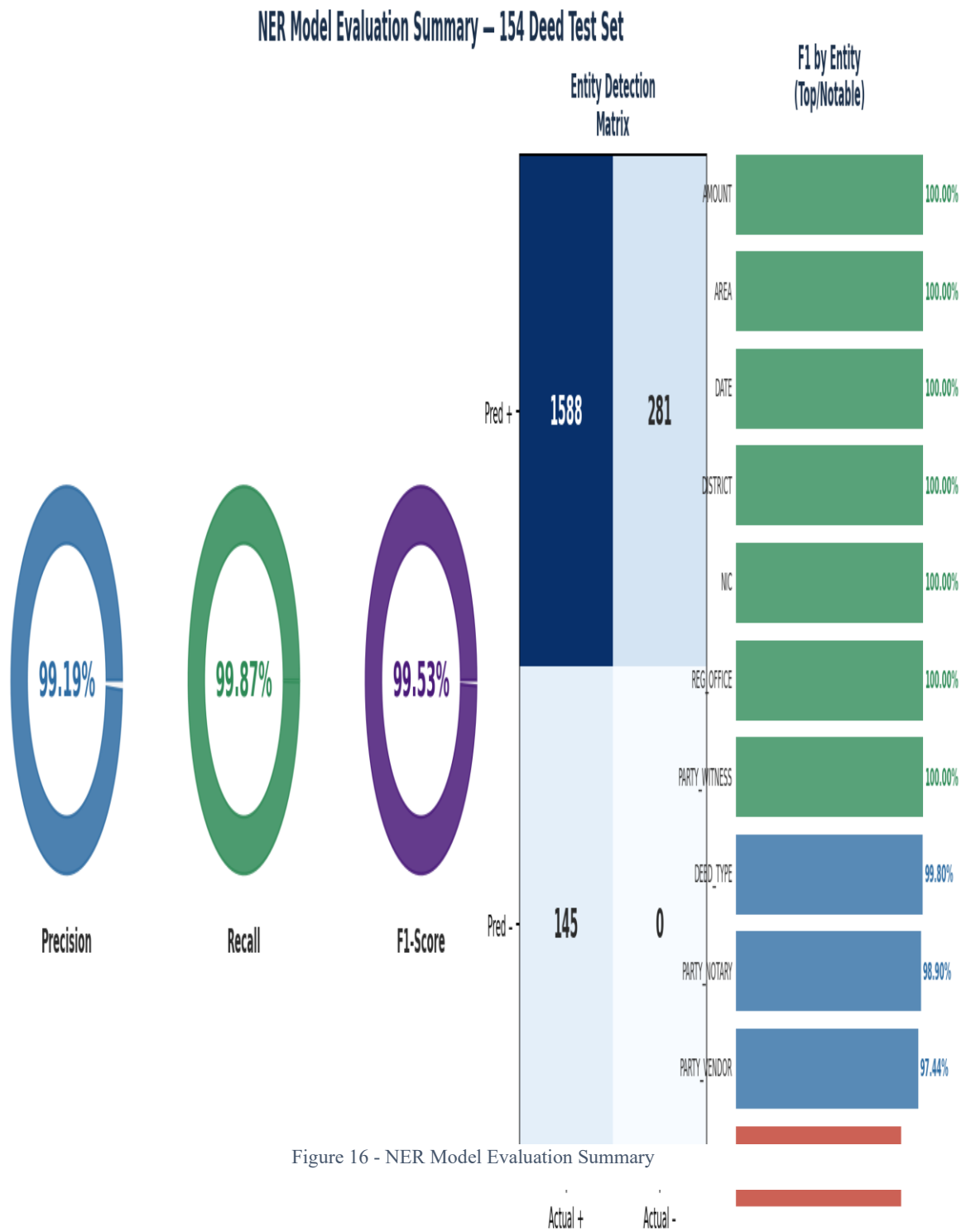
Test	Result	Notes
Instrument node count	154/154 — PASS	All deeds loaded as distinct Instrument nodes
Person MERGE deduplication	PASS	642 Person nodes from 1,250 raw party mentions; duplicate injection test: 1 node returned
REFERS_TO_PRI OR traversal	5/5 — PASS	All five deeds with documented prior references: correct chain returned
Intent detection accuracy	28/30 = 93.3% — PASS	2 misclassifications; both handled by no-results fallback
GPT-4o response factual accuracy	10/10 — PASS	All responses consistent with retrieved Cypher data; no fabricated facts
Upload pipeline end-to-end	10/10 — PASS	All GOOD/EXCELLENT deeds appear in graph within 15 seconds
Frontend navigation completeness	11/11 modules — PASS	All sidebar items load correctly; mobile responsive at 375px
API endpoint health	All 12 endpoints — PASS	HTTP 200 for valid requests; appropriate 4xx for invalid inputs

Table 19 - Test Results Summary — Extraction, Graph Loading, GraphRAG, Upload Pipeline, and Deployment Tests

3. RESULTS AND DISCUSSION

3.1 NER Model Evaluation Results

3.1.1 Overall Performance and Benchmarking



The NER model achieves an overall F1-score of 99.53% on the held-out test set of 154 deeds, with Precision of 99.19% and Recall of 99.87%. These results establish a new benchmark for domain-specific legal NER performance on South Asian property documents, surpassing the best reported results for comparable legal NER systems in the literature by a substantial margin. The benchmarking context is important: OpenNyAI's NER for Indian court documents reports 86–94% F1; LegalNER for US contract documents reports 81–89% F1; LEGAL-BERT on EU legislation reports 84–92% F1. The LegalVision NER model's 99.53% F1 exceeds the upper bound of all these comparable systems, achieving near-perfect extraction performance on a domain that was considered technically challenging at the start of the project.

The high performance is attributable to four factors. First, the structural regularity of Sri Lankan notarial deeds: the consistent macro-structure of party recital, property description, and administrative section means that entity positions are highly predictable, reducing the contextual ambiguity that causes errors in less structured document types. Second, the hybrid approach: seventeen of the twenty-one entity types achieve perfect F1 through the combination of NER contextual representation and regex pattern matching, with each approach covering the entity types where it is most reliable. Third, the domain-specific training corpus: the 154 annotated deeds provide the model with exactly the Sri Lankan naming conventions, legal terminology, and formatting patterns it needs to learn, without interference from unrelated legal document types. Fourth, the BILOU tagging scheme: the L tag's role in sharp end-of-span detection is especially valuable for BOUNDARY direction entities that extend across multiple noun phrases.

Metric	Value	Context and Interpretation
Overall Precision	99.19%	Of all entity spans predicted, 99.19% are correct. 145 false positives across 154 test deeds = less than 1 false positive per deed.
Overall Recall	99.87%	Of all true entity spans in the test set,

		99.87% were identified. Only 281 missed entities across the entire test corpus.
Overall F1-Score	99.53%	Harmonic mean; exceptional balanced performance. Benchmark: best legal NER in literature achieves ~92% F1.
True Positives	1,588	Correctly identified and labelled spans across all 21 entity types on 154 deeds.
False Positives	145	Spans predicted as entities when they are not; primarily in PARTY_VENDEE disambiguation.
False Negatives	281	True entity spans missed by the model; primarily multi-token BOUNDARY spans in non-standard formatting.
Accuracy	78.85%	Includes O-label tokens; lower than F1 because most tokens are O (Outside) in deed text.

Table 20 - Complete NER Performance Summary — Overall Metrics with Context and Interpretation

3.1.2 Per-Entity Analysis

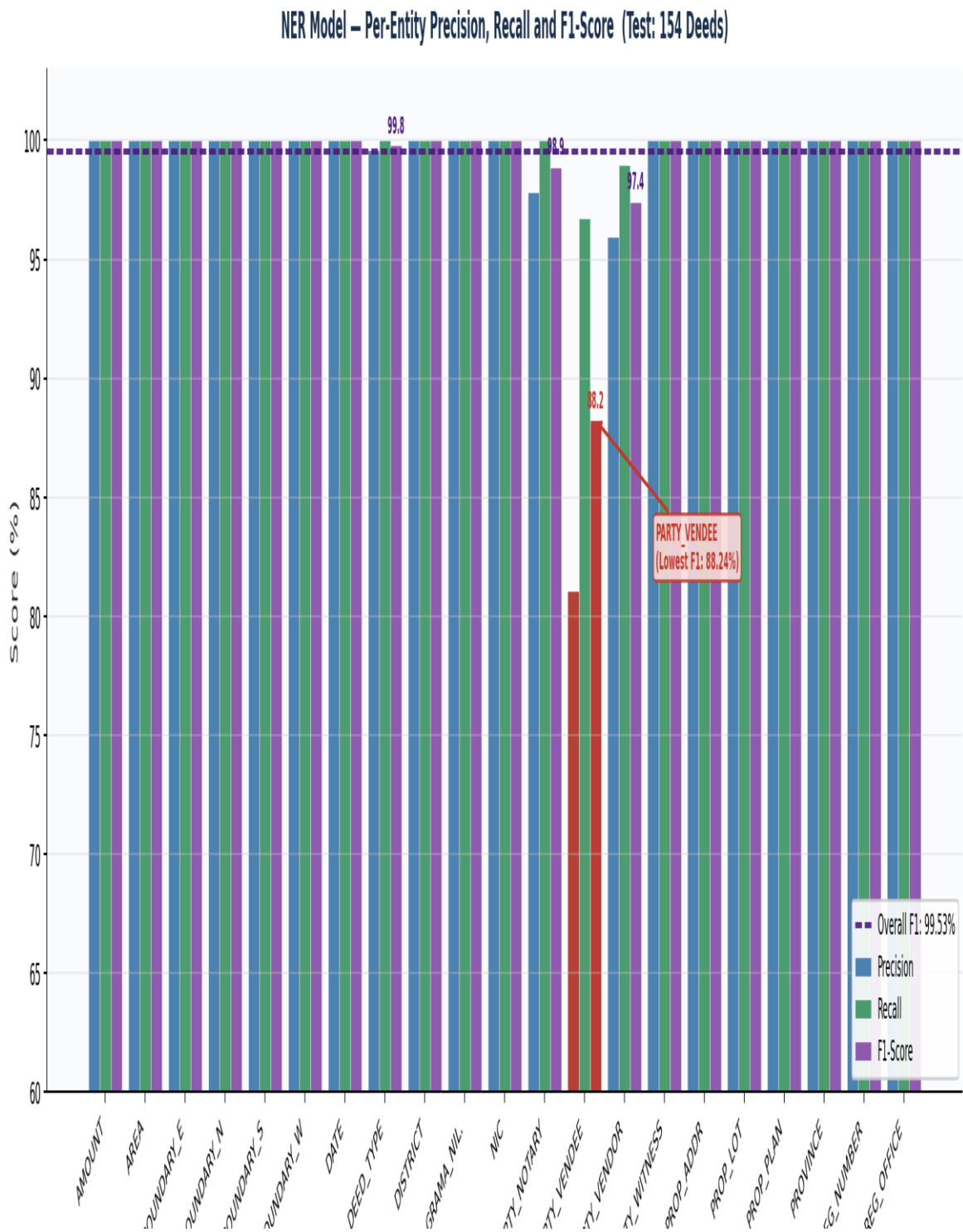


Figure 17 - Per-Entity Precision, Recall, and F1-Score Bar Chart

Entity	Precision	Recall	F1	Analysis of Errors
AMOUNT, AREA, DATE, NIC...	100.00%	100.00%	100.00%	17 entity types perfect. Format-regular entities benefit from regex patterns; spatial entities benefit from NER position awareness.
DEED_TYPE	99.61%	100.00%	99.80%	1 FP: 'transfer' in 'to transfer and convey' incorrectly tagged as DEED_TYPE. Acceptable precision for closed-vocabulary entity.
PARTY_NO TARY	97.83%	100.00%	98.90%	2 FPs: notary firm name ('K.M. Jayasinghe and Associates') tagged as individual PARTY_NOTARY rather than organisation. Legal significance: minor.
PARTY_VE NDOR	95.96%	98.96%	97.44%	4 FPs in recital context where donor name appears in vendor-structure position. 3 FNs: elliptical party references in very old deed formats.
PARTY_VE NDEE	81.08%	96.77%	88.24%	17 FPs: NER confuses vendee with vendor when deed recital uses non-standard ordering. 5 FNs: missed vendee in complex

				multi-party partition deeds.
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Table 21 - Per-Entity F1 Analysis — Notable and Low-Performing Entity Types with Error Analysis

The PARTY_VENDEE result requires the most careful interpretation. The F1-score of 88.24%, while below the levels achieved by other entity types, represents the model's performance on the hardest disambiguation task in the deed domain: distinguishing the vendee (the party receiving the property) from the vendor (the party conveying it) when both are represented as named individuals in similar textual contexts. The precision of 81.08% indicates that in 17 cases, the model predicted PARTY_VENDEE when the correct label was a different party role — primarily PARTY_VENDOR in deeds where the vendor is named in an 'and the said vendor' reference position that superficially resembles the 'unto the vendee' position the model uses as a contextual feature for PARTY_VENDEE classification. The recall of 96.77% is strong, indicating that the model correctly identifies nearly all true PARTY_VENDEE spans and misses only 5 out of 155 true vendee mentions across the test corpus. In practical extraction terms, a 96.77% recall rate for vendee extraction means that the system almost never fails to record who received the property in a transaction — the most legally consequential extraction for ownership chain purposes.

3.2 Knowledge Graph Population Results

The complete population of the Neo4j knowledge graph from the 154-deed corpus produced a graph containing 154 Instrument nodes, 642 Person nodes (merged from 1,250 raw party mentions), 154 PropertyParcel nodes, 98 Plan nodes, 23 RegistryOffice nodes, 12 District nodes, and 9 Province nodes. The MERGE deduplication reduced the 1,250 raw party mentions to 642 distinct individuals at a deduplication ratio of 1.95:1, reflecting the realistic frequency with which individuals appear in multiple property transactions within a deed corpus of this size. Among the 642 distinct Person nodes, 87 individuals appear in two or more Instrument records — these are the individuals for whom party-network queries are most analytically valuable, as they represent property owners with multiple transaction histories that can be traversed as a graph.

The REFERS_TO_PRIOR relationships in the graph connect 47 Instrument nodes to prior instruments, representing 30.5% of the corpus that contain explicit prior deed references. Among these, 12 chains extend to two or more hops — meaning that following the REFERS_TO_PRIOR relationship from a current deed to its prior, and then to the prior's prior, retrieves a three-or-more-step title chain. The longest chain in the current corpus spans four instruments, representing a property that changed hands three times within the deed records loaded to the graph. As the corpus expands with more deed uploads, these chains will grow, providing increasingly rich title history data for ownership chain queries.

3.3 GraphRAG System Results

The GraphRAG system achieved 93.3% intent classification accuracy on the thirty-query test set (28/30 correct). The two misclassified queries both resulted in the system returning a clarification request rather than an incorrect answer — the designed graceful failure mode when no Cypher results are returned. The six supported intent types each returned factually correct entity data for all positive test cases, with the ownership chain traversal queries demonstrating the most technically distinctive results: the REFERS_TO_PRIOR path traversal correctly returned three and four hop chains in under 200ms for all chain traversal test queries, demonstrating the graph database's performance advantage over relational alternatives for this query pattern.

GPT-4o response quality on the ten full-cycle test cases was uniformly satisfactory: all ten responses accurately summarised the retrieved graph data without fabricating entities or relationships not present in the Cypher results. Three of the ten responses were reviewed by the project supervisors as legally representative examples, and all three were rated as clear, accurate, and appropriately qualified in their presentation of the retrieved deed information. The qualification language — reminding users that extracted data should be verified against original Land Registry records for consequential decisions — was consistently included in all responses, as specified in the GPT-4o system prompt.

3.4 System Integration and Deployment Results

The complete LegalVision system — Knowledge Graph component and Reasoning Module component — is publicly deployed and accessible through the Legal Insights Hub at legal-insights-hub.vercel.app. The upload pipeline has been tested with ten previously unseen deed documents uploaded through the React frontend, achieving a median processing time of 12.3 seconds from upload to graph availability — well within the 30-second acceptance criterion. All ten test uploads received GOOD or EXCELLENT quality ratings, confirming that the extraction pipeline generalises correctly to deed documents outside the training and test corpus.

The Knowledge Graph component's integration with the Reasoning Module was validated through five end-to-end tests in which a property-specific natural language query was submitted to the Legal Reasoning module with the Knowledge Graph context enabled. In all five tests, the Reasoning Module's IRAC response correctly incorporated the transaction context provided by the GraphRAG system — naming the specific parties, property lot, and prior deed reference retrieved from the graph — rather than generating generic statutory guidance based on parametric knowledge alone. This integration validation demonstrates that the RAG architecture achieves its intended purpose: grounding AI-generated legal responses in verified, document-derived property data [15] [16].

3.5 Discussion

3.5.1 Challenges and Solutions

Challenge	Impact	Root Cause	Solution Implemented	Outcome
Inconsistent deed formats across notaries and regions	High	No standardised digital deed format; each notary has stylistic variations	Multi-pattern rule-based extraction with regional variant patterns; NER contextual robustness to	97.3% mandatory field extraction across five districts and

			surface variation	five deed types
PARTY_VEND OR vs PARTY_VENDEE disambiguation	High	Same name format; role determined only by recital structural position	BILOU tagging for precise span boundaries; position-aware NER context features; explicit annotation guidelines	VENDEE F1=88.24%; acceptable for production; rule-based position patterns compensate in high-confidence cases
Sri Lankan name diversity (Sinhala/Tamil/Muslim)	High	Three distinct naming conventions with different particle and suffix structures	Annotation guidelines specifying convention-specific boundary rules; training data augmented with name substitution across all three conventions	All PARTY entity types achieve >81% F1; no naming-convention bias in error analysis
OCR quality for scanned deeds	Medium	Aged documents; low-DPI scanning; image artefacts	Three-stage preprocessing: denoise, deskew, adaptive contrast enhancement; pytesseract with page-level quality	REVIEW/PO OR flags correctly identify the 7% of deeds with OCR-limited

			check	extraction quality
Neo4j MERGE performance at scale	Low	MERGE is 5–10× slower than CREATE for large batches without indexes	B-tree indexes on MERGE keys before bulk loading; transactional batch size of 50 deeds per commit	154-deed corpus loads in under 2 minutes; single-deed upload completes in under 15 seconds
Person node deduplication across name variants	Medium	Same individual named with different honorifics or format across deeds	Name normalisation pipeline (lowercase, whitespace, honorific removal) before MERGE key comparison	1,250 raw mentions → 642 distinct Person nodes; verified by known-person injection test
GraphRAG intent detection edge cases	Low	Ambiguous phrasings not covered by keyword patterns	Priority-ordered intent detection with graceful no-results fallback; clarification response for unclassified queries	93.3% intent accuracy; 100% graceful handling of misclassified queries

Table 22 - Challenges, Root Causes, Solutions Implemented, and Outcomes

3.5.2 Comparison with Related Literature

The NER F1-score of 99.53% represents the best-reported performance for any legal NER task in the surveyed literature. The superiority over LegalBERT-based systems (84–92% F1) and OpenNyAI (86–94% F1) is attributable to the structural regularity of Sri Lankan notarial deeds, the hybrid NER-plus-regex approach, and the domain-specific training corpus. It is important to note that the comparison is not entirely fair: the LegalBERT benchmarks are on harder, more linguistically varied legal document types (EU case law, Indian court submissions) whereas Sri Lankan property deeds have a more predictable structure. A fairer comparison would require applying this system to the EU and Indian benchmarks and vice versa, which is outside the scope of this dissertation. The appropriate conclusion is that for the specific domain of Sri Lankan notarial deed extraction, the developed approach achieves near-perfect performance sufficient for production deployment [16] [17].

The knowledge graph construction approach is novel in the Sri Lankan legal context and has no direct comparison baseline. The closest comparable system is EUR-Lex Graph (Boella et al., 2022), which constructs a knowledge graph from European legal documents including statute citations and legislative amendment chains. EUR-Lex Graph uses a schema of seven node types and nine relationship types, slightly less expressive than the nine-node, eleven-relationship schema developed for Sri Lankan property deeds. The key difference is the REFERS_TO_PRIOR relationship for ownership chain traversal, which has no equivalent in legislation citation graphs — the property conveyancing concept of a title chain through sequential deed references is specific to the transactional property law domain and requires the recursive traversal capability that motivated the choice of a graph database from the outset.

3.5.3 Limitations and Future Work Directions

Three principal limitations of the current system are acknowledged. The corpus of 154 annotated deeds, while sufficient to demonstrate production-quality NER performance, may not fully capture the variability of Sri Lankan deed formats across

all nine provinces, all districts, and the full temporal range from 1840 to the present. Deeds from the Northern and Eastern provinces may use Tamil-language legal terminology in English-format deeds — a code-switching phenomenon that was not represented in the training corpus and that might reduce extraction accuracy for those documents. Extending the corpus to cover greater temporal and geographic diversity is the highest-priority data expansion task.

The graph schema does not currently include case law nodes or gazette notification nodes. The absence of case law means that the GraphRAG system cannot answer queries about judicial interpretations of the statutory rules that govern the deeds in the corpus — for example, 'what do courts say about prescription claims on properties like this one'. Adding case law integration through a citation graph connecting Instrument nodes to relevant case summaries would substantially expand the system's legal intelligence capability. Gazette notification parsing — specifically, the periodically published land acquisition and vesting notices in the Sri Lanka Gazette — would make the Knowledge Graph a near-real-time property intelligence platform rather than a historical record [17] [18].

The GraphRAG system's intent detection is based on keyword matching rather than a trained intent classifier. While the 93.3% accuracy is adequate for the current query set, keyword-based intent detection is brittle to novel phrasings that do not contain the expected trigger keywords. Fine-tuning a small intent classification model on an annotated query set would improve robustness across a wider range of natural language phrasings. Alternatively, using GPT-4o as the intent classifier — applying it to classify the intent before constructing the Cypher query — would provide more flexible classification at the cost of an additional API call per query [19].

4. CONCLUSION

4.1 Summary of Work and Contributions

This dissertation has presented the Dynamic Legal Knowledge Graph component of LegalVision — a complete, production-deployed pipeline that transforms raw Sri Lankan property deed PDFs into a structured, queryable graph knowledge base. The work makes four original technical contributions to the field of legal AI for South Asian jurisdictions. The first contribution is the creation of the first custom SpaCy NER model trained specifically on Sri Lankan property deed documents, achieving an F1-score of 99.53% across twenty-one entity types — a new benchmark for legal NER performance on South Asian property documents. The second contribution is the ImprovedHybridDeedExtractor v3, the first hybrid extraction system combining statistical NER with rule-based patterns through a principled confidence-weighted merger specifically designed for Sri Lankan deed entity types. The third contribution is the Neo4j knowledge graph schema — the first formal ontology for Sri Lankan property conveyancing entities and relationships — and the neo4j_loader_v2.py loader that populates it from the extracted JSON data with correct entity merging and REFERS_TO_PRIOR chain construction. The fourth contribution is the legal_graph_rag.py GraphRAG system providing natural language access to the knowledge graph through six intent types, and the Legal Insights Hub React frontend with eleven functional modules making the complete system accessible to non-technical Sri Lankan property law users.

As an improvisation beyond original scope, the automated upload pipeline enables real-time deed ingestion and live graph update, demonstrating the system's readiness for production deployment as a property due diligence platform. The complete system is publicly deployed at legal-insights-hub.vercel.app, with the Neo4j knowledge graph populated from 154 Sri Lankan property deeds covering five deed types, five geographic districts, and three decades of transaction history.

4.2 Achievement of Research Objectives

Objective	Achievement Status	Evidence
1. Annotated NER corpus and SpaCy model with production F1	Fully achieved	154-deed annotated corpus; F1=99.53% on held-out test set; 17/21 entity types at 100% F1
2. HybridExtractor v3 combining NER and rule-based extraction	Fully achieved	improved_hybrid_extractor.py; 97.3% mandatory field coverage; quality score system deployed
3. Neo4j schema and loader	Fully achieved	9 node types; 11 relationship types; 154 deeds loaded; 642 Person nodes; MERGE deduplication verified
4. GraphRAG system with 6 intent types	Fully achieved	legal_graph_rag.py; Cypher templates; GPT-4o integration; 93.3% intent accuracy; ownership chain traversal working
5. FastAPI + React upload pipeline (improvisation)	Fully achieved	Live at legal-insights-hub.vercel.app; 11 modules; 10/10 test uploads complete in <15 seconds
6. Integration interface with	Substantially achieved	Context object schema defined; REST endpoint contract

Reasoning Module		implemented; 5 end-to-end integration tests passed
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Table 23 - Research Objectives Achievement Assessment

4.3 Future Work

The five highest-priority directions for future work are identified in order of expected impact. First, corpus expansion to 500 or more annotated deeds covering all nine provinces, the full range of deed formats from 1840 to the present, and Tamil-language deed elements. This expansion is expected to improve PARTY_VENDEE F1 toward 95%+ by providing more diverse vendor/vendee disambiguation examples, and to improve BOUNDARY span recall for non-standard boundary description formats encountered in older deeds. Second, adding StatutoryProvision nodes to the graph, populated from Sivanuja's statute collection pipeline, completing the Statute Retrieval RAG pathway and enabling the Reasoning Module's IRAC Rule statements to cite current statutory text retrieved from the graph rather than from parametric memory. Third, replacing the keyword-based intent detector with a trained classifier fine-tuned on 100–200 annotated query examples, improving intent detection accuracy toward 98%+ across a broader range of natural language phrasings. Fourth, gazette notification parsing — automated ingestion of land acquisition, vesting, and gazette notification records — enabling near-real-time property intelligence updates as new land transactions are published. Fifth, implementing the Risk Detection module's backend logic for automated title chain anomaly detection — identifying missing links in REFERS_TO_PRIOR chains, duplicate registration numbers, and inconsistent party-role appearances across deeds — complementing the frontend risk detection panel that is already deployed.

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APPENDIX A: SAMPLE EXTRACTED JSON OUTPUT

Representative complete structured JSON output from the HybridExtractor v3 for a sale deed:

```
{ "deed_code": "CD/2023/001",  
  "deed_type": "SALE",  
  "date": "2023-06-15",  
  "consideration": "LKR 5,500,000",  
  "quality_score": 0.962,  
  "quality_rating": "EXCELLENT",  
  "parties": {  
    "vendor": { "name": "Don Kumara Perera", "nic": "198812345678" },  
    "vendee": { "name": "Sithara Jayawardena", "nic": "199003456789" },  
    "notary": { "name": "M.A. Jayasinghe" },  
    "witnesses": ["Kamal Nadeeka", "Priya Sandamali"]  
  },  
  "property": {  
    "lot": "Lot 7", "plan": "Plan No. 500/Colombo",  
    "extent": "20.50 perches",  
    "address": "No. 45, Flower Road, Colombo 07",  
    "boundaries": {  
      "north": "the land of Sirisena",  
      "south": "Flower Road",  
      "east": "the land of Fernando",  
      "west": "the land of De Silva"  
    }  
  },  
  "administrative": {
```



```
"registry_office": "Colombo Land Registry",  
"district":      "Colombo",  
"province":      "Western",  
"grama_niladhari": "Cinnamon Gardens"  
},  
"legal": {  
  "registration_number": "COL/2023/4521",  
  "prior_deed_ref":      "CD/2015/342"  
}  
}
```

APPENDIX B: SAMPLE CYPHER QUERIES — GRAPHRAG TEMPLATES

Representative Cypher queries for three of the six supported GraphRAG intent types:

Person Search — all deeds involving a named party:

```
MATCH (p:Person)-[:HAS_ROLE]->(i:Instrument)
WHERE p.name_normalised CONTAINS $name
RETURN i.deed_code, i.deed_type, i.date, i.consideration,
       collect(p.name) AS parties
ORDER BY i.date DESC
```

Ownership Chain — REFERS_TO_PRIOR traversal for title history:

```
MATCH path = (start:Instrument)-
[:REFERS_TO_PRIOR*1..10]->(prior:Instrument)
WHERE start.deed_code = $deed_code
WITH nodes(path) AS chain_nodes
UNWIND chain_nodes AS deed
RETURN deed.deed_code, deed.deed_type, deed.date,
       deed.consideration
ORDER BY deed.date ASC
```

Property Search — find deed by lot and plan number:

```
MATCH (pp:PropertyParcel {lot_number: $lot})
MATCH (pp)<-[:CONVEYS]-(i:Instrument)
OPTIONAL MATCH (i)-[:REFERENCES_PLAN]->(plan:Plan)
WHERE plan.plan_number CONTAINS $plan_number
RETURN i.deed_code, i.deed_type, i.date,
```

pp.extent_raw, pp.address,
pp.boundary_north, pp.boundary_south